

Introduction to dashboards

DASHBOARD DESIGN CONCEPTS



Olga Scrivner, PhD
President, Scrivner Solutions Inc

Data deluge



2.5 million terabytes data every day

¹ Connall, M. (2020). Top 20 big data facts and statistics for 2022.

Data deluge



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1.5 MB of data created for every person every second

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79 zettabytes of data generated worldwide in 2021

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Data deluge



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Most companies only analyze 12% of the data

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Transforming data into actionable insights

Turning data into information, and information into actionable insights

Analytics

The critical technology that brings
value out of data

Transforming data into actionable insights

Turning data into information, and information into actionable insights

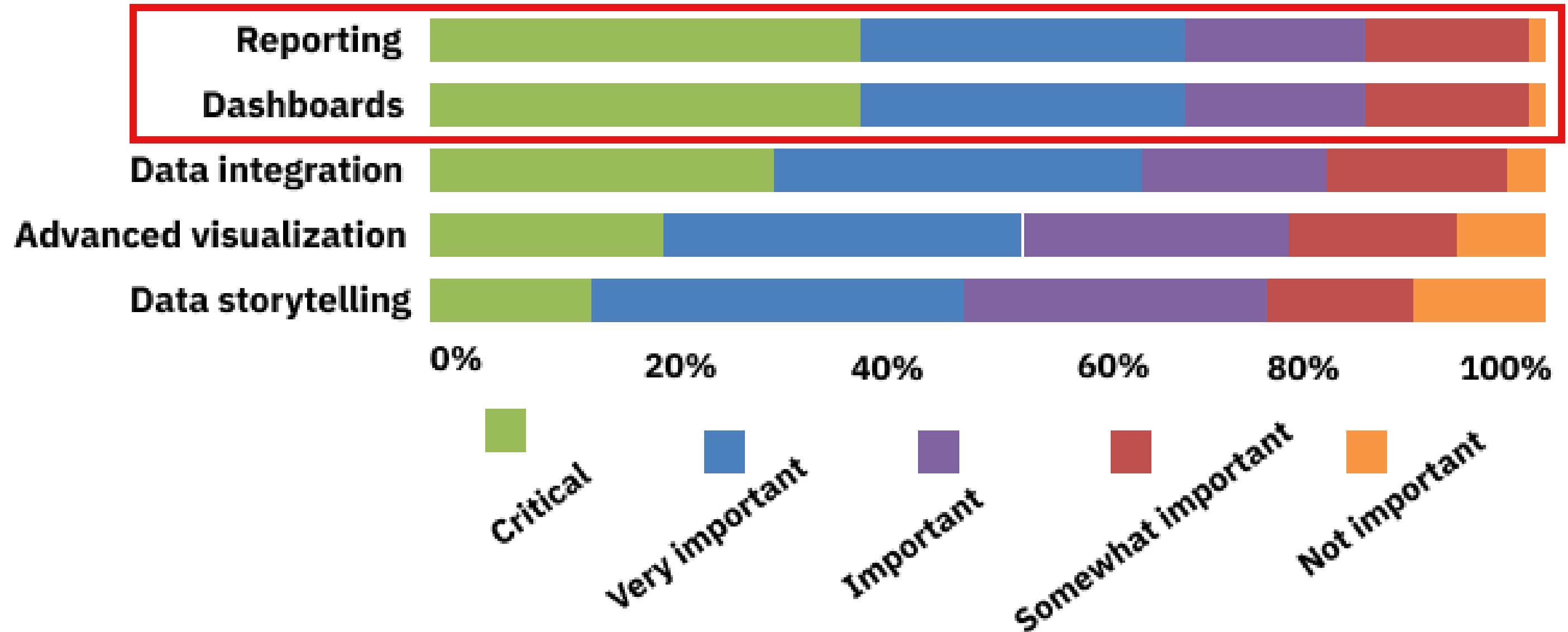
Analytics

The critical technology that brings
value out of data

**Visual
Analytics**

Values are translated into
actionable insights

Visual analytics technologies



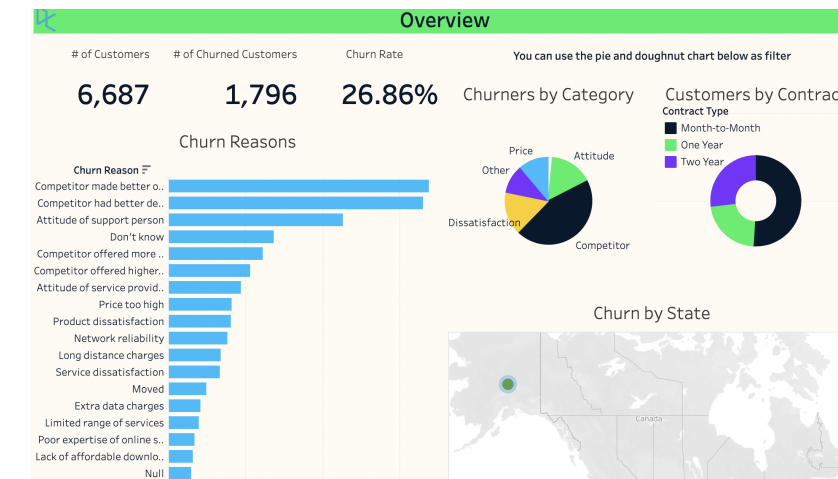
Reports versus dashboards

Report

Social Media	KPI 1	KPI 2	KPI 3
Impressions	526	829	1,231
Growth	-	58.21%	-31.31%
Accounts	112	253	422
Growth	-	125.89%	-35.35%

- Static snapshot in time
- Multiple pages
- Detailed historical data
- Tables and basic graphics

Dashboard



- Dynamic views
- Single page
- Actionable information
- Interactive graphics

Check the dashboard!

A visual display of the most important information needed to achieve one or more objectives, arranged on a single screen so the information can be **monitored at a glance**.¹

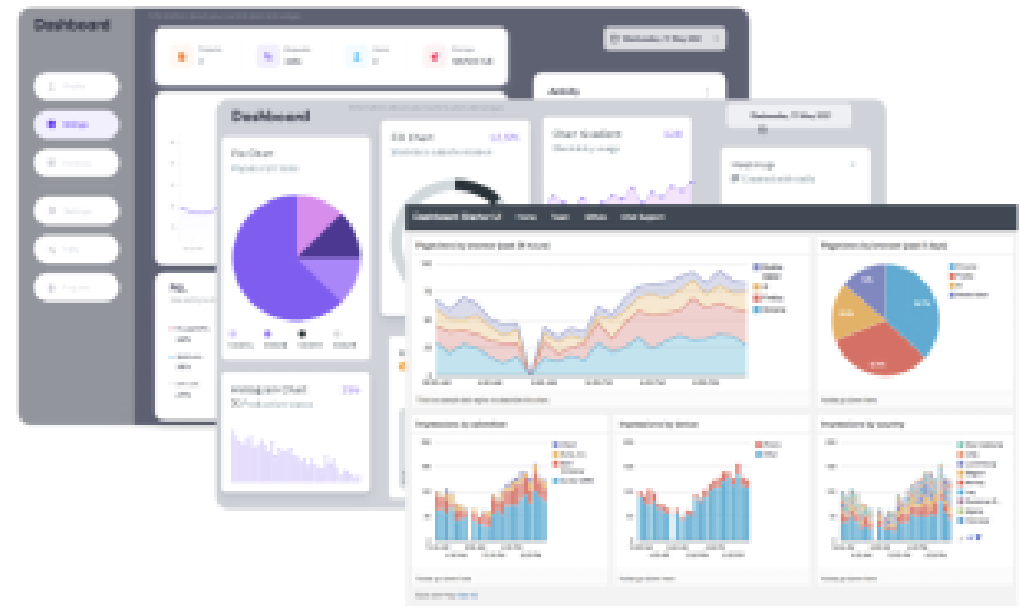


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A visual display of the most important information needed to achieve one or more objectives, arranged on a single screen so the information can be monitored at a glance.¹

Increased Efficiency



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Increased Efficiency

Improved Processes



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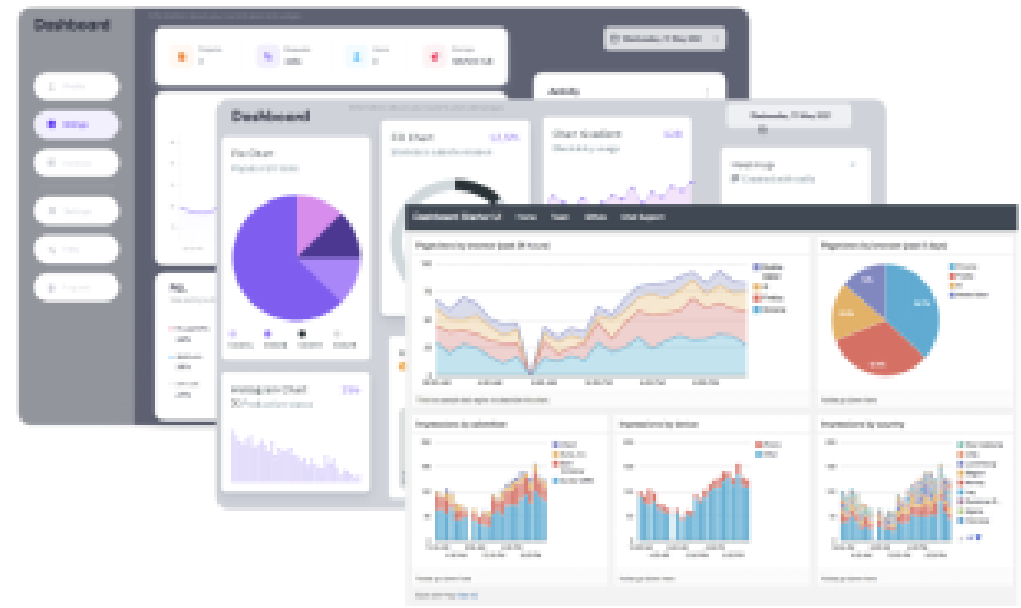
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Increased Efficiency

Improved Processes

Operational & Financial
Insights



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Return on Investment



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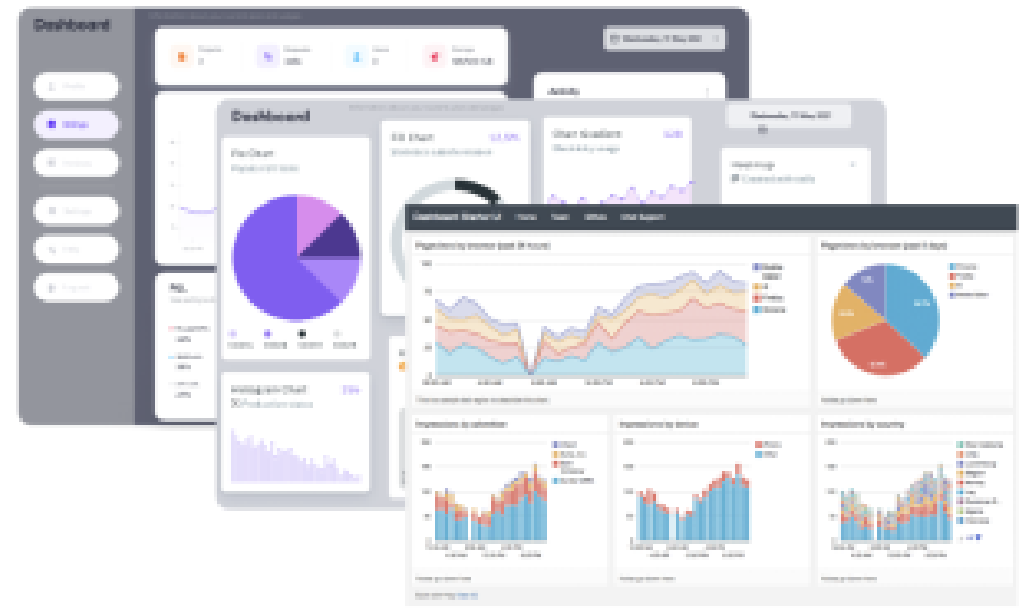
Increased Efficiency

Improved Processes

Operational & Financial
Insights

Return on Investment

Increased Productivity



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Check the dashboard!

A visual display of the most important information needed to achieve one or more objectives, arranged on a single screen so the information can be monitored at a glance.¹



What's happening? What's happened? What's going to happen? Check the dashboard!²

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What is an actionable dashboard?

Actionable dashboard informs business decisions



Including the right KPIs

¹ Mellisa King. (2022). How to Create Actionable Dashboards: 5 Best Practices.

What is an actionable dashboard?

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Including the right KPIs



Providing accurate information

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Including the right KPIs



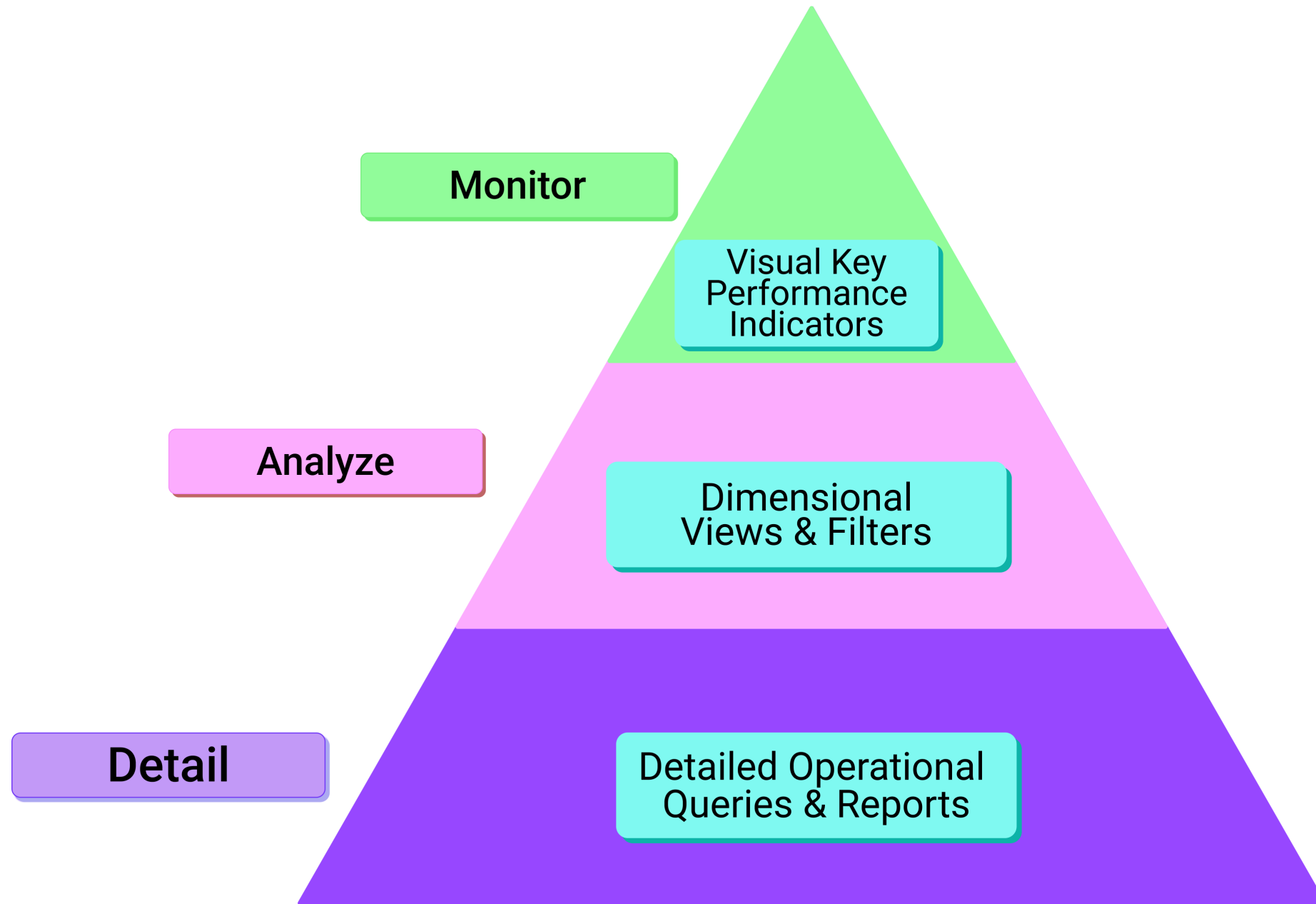
Providing accurate information



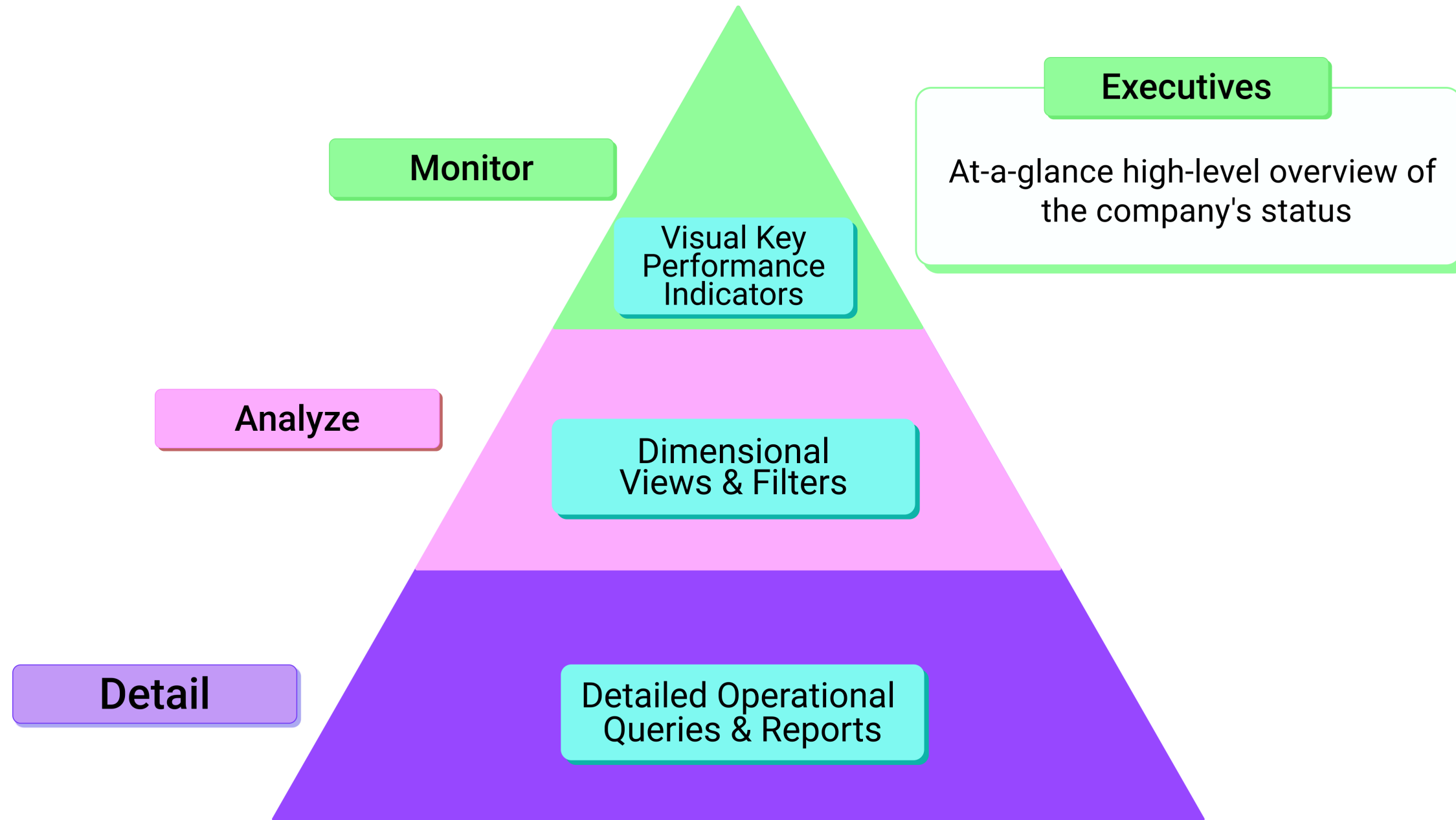
Defining the right audience

¹ Mellisa King. (2022). How to Create Actionable Dashboards: 5 Best Practices.

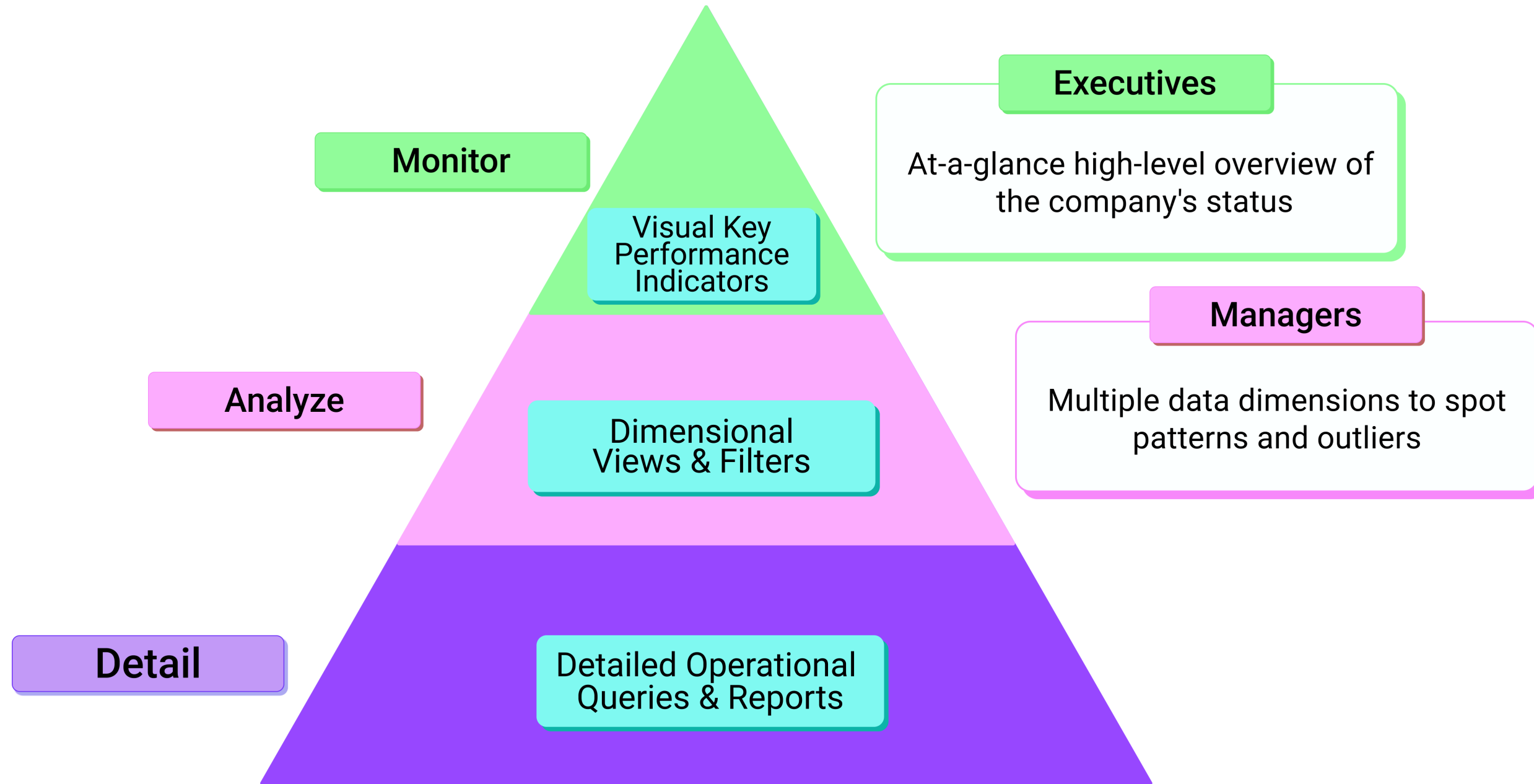
The MAD framework: monitor-analyze-detail



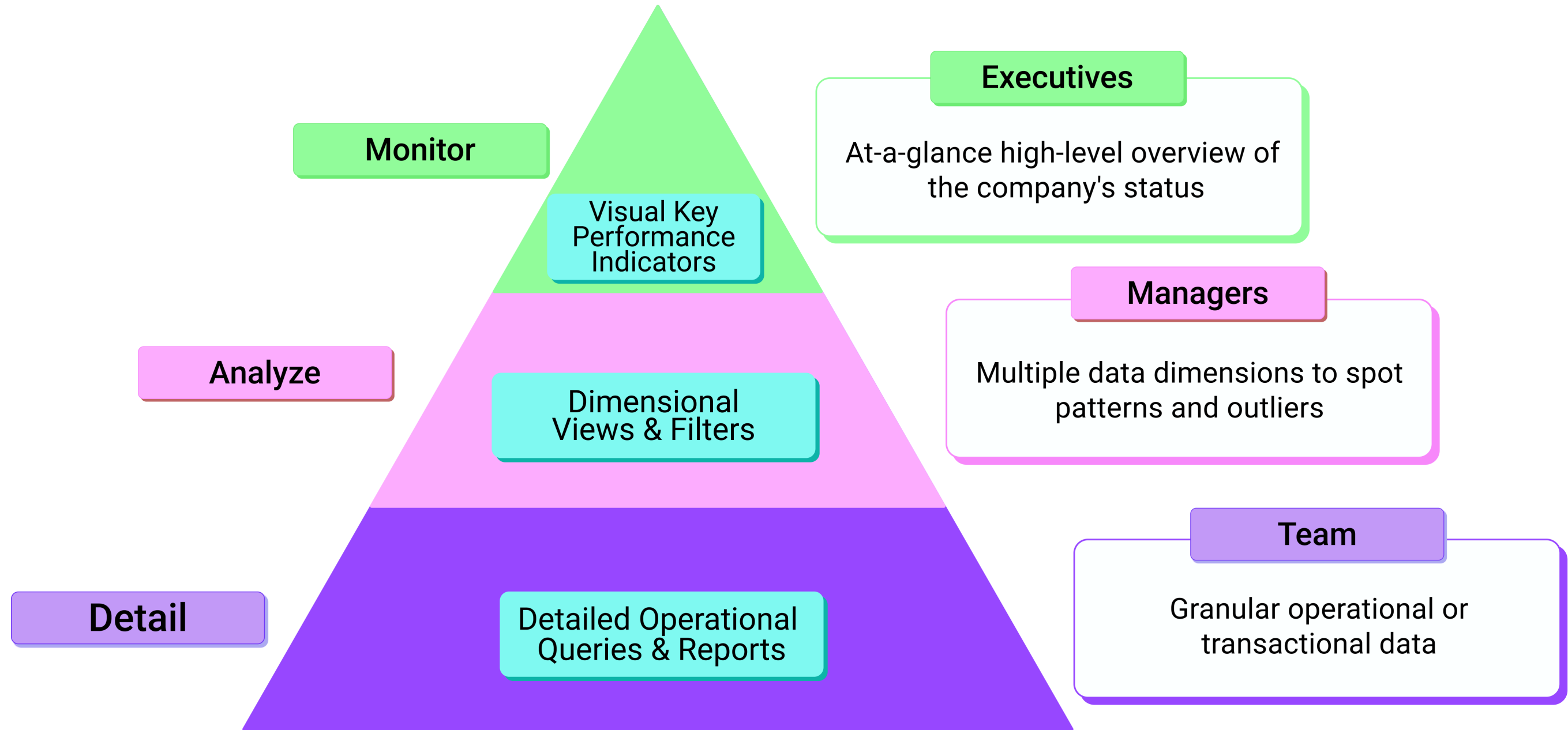
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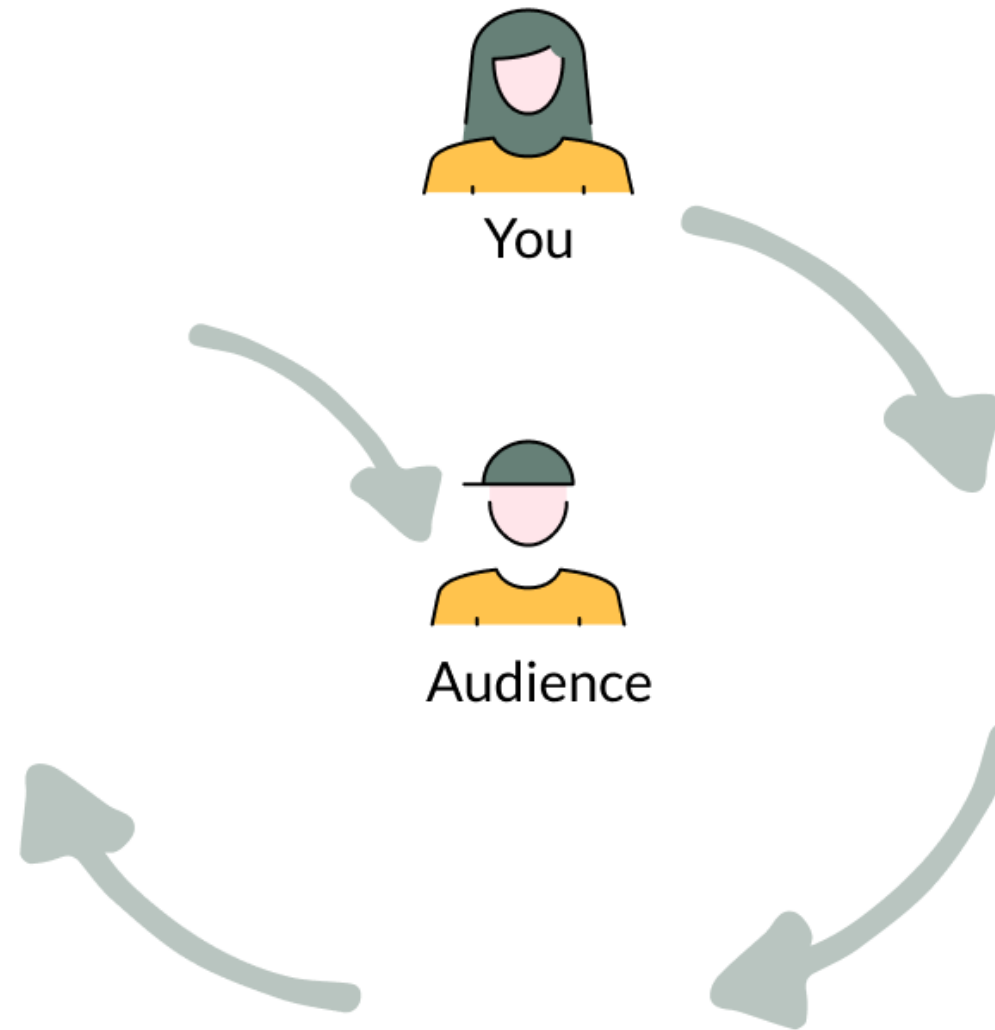
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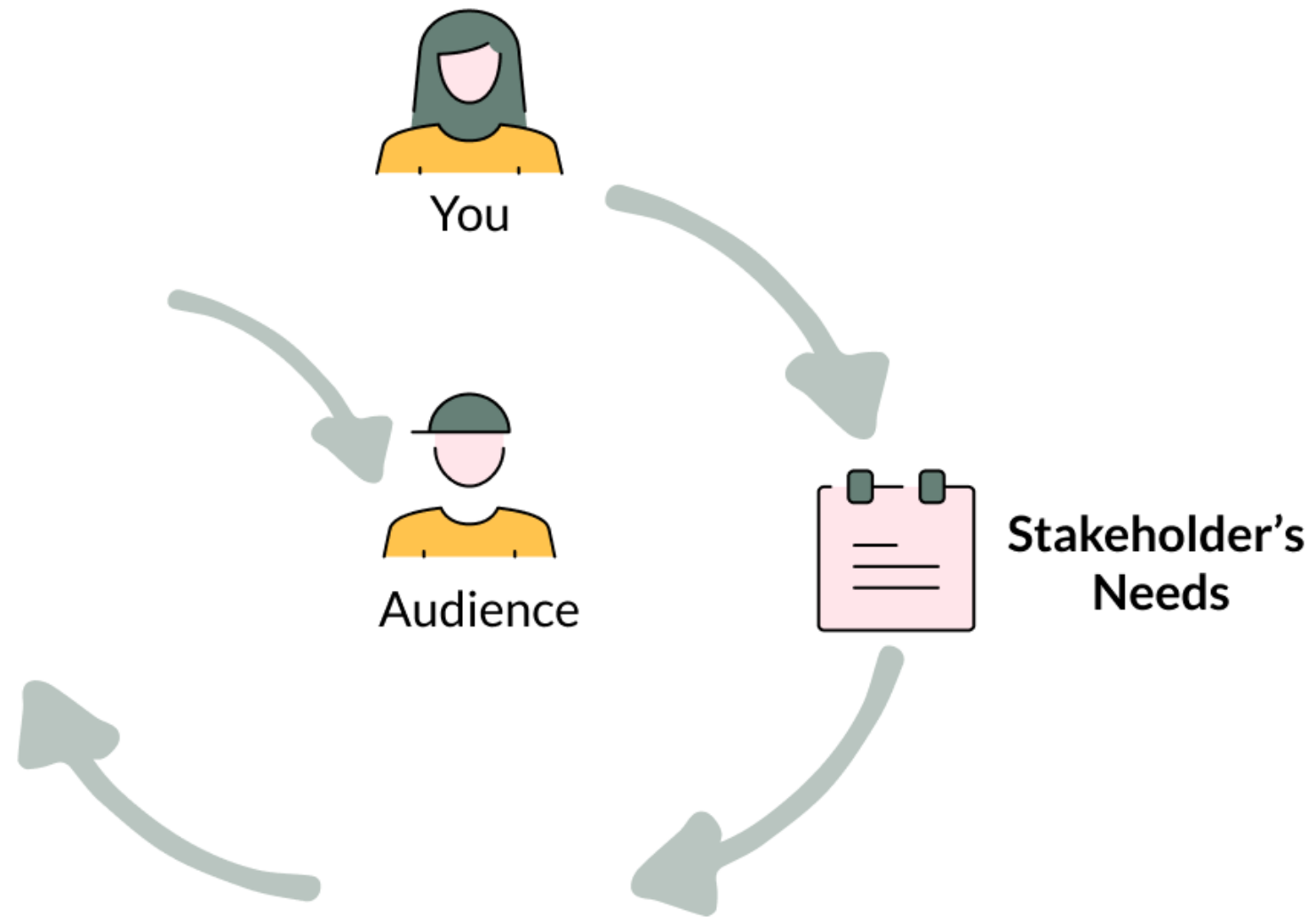
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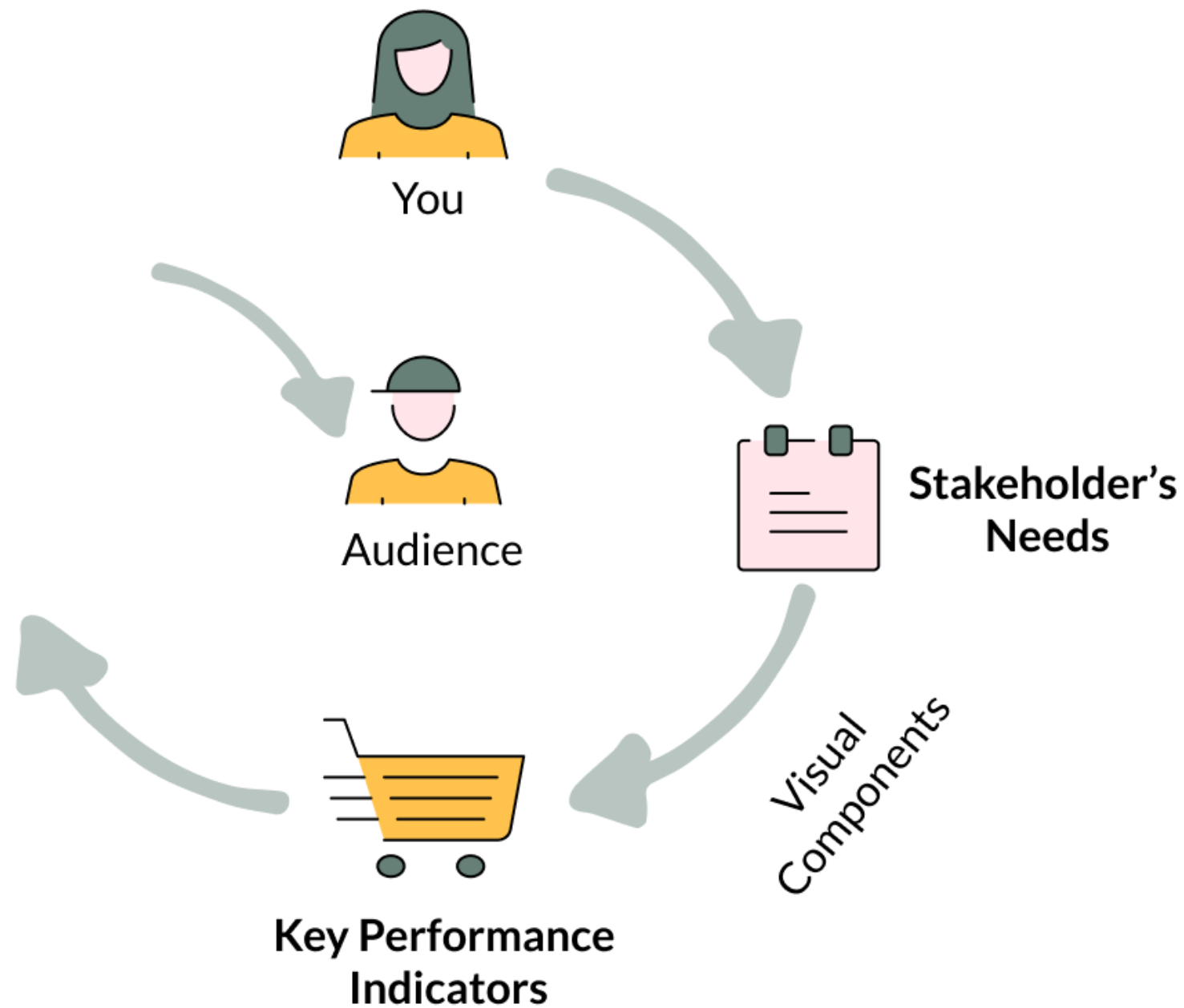
Course overview



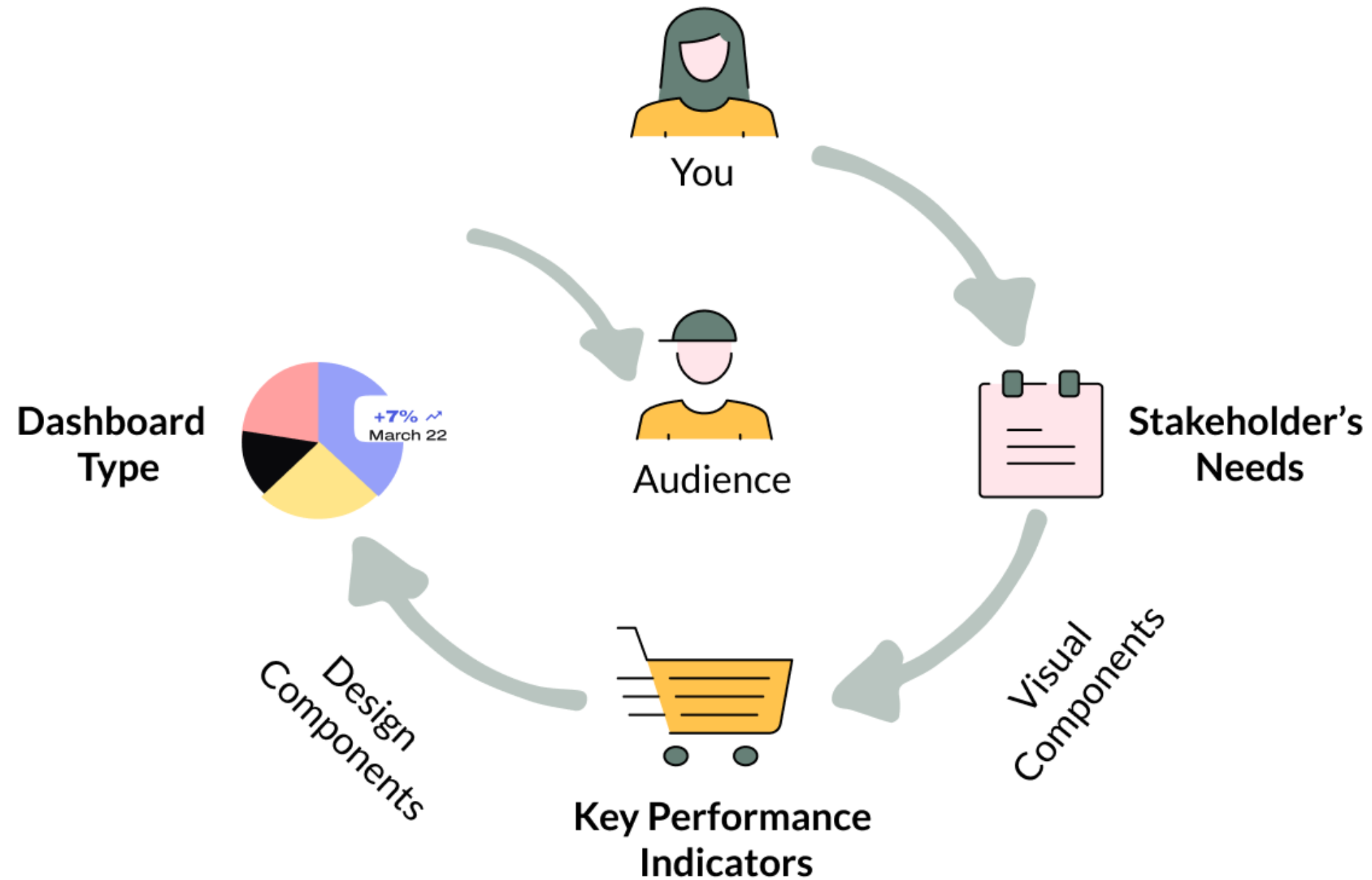
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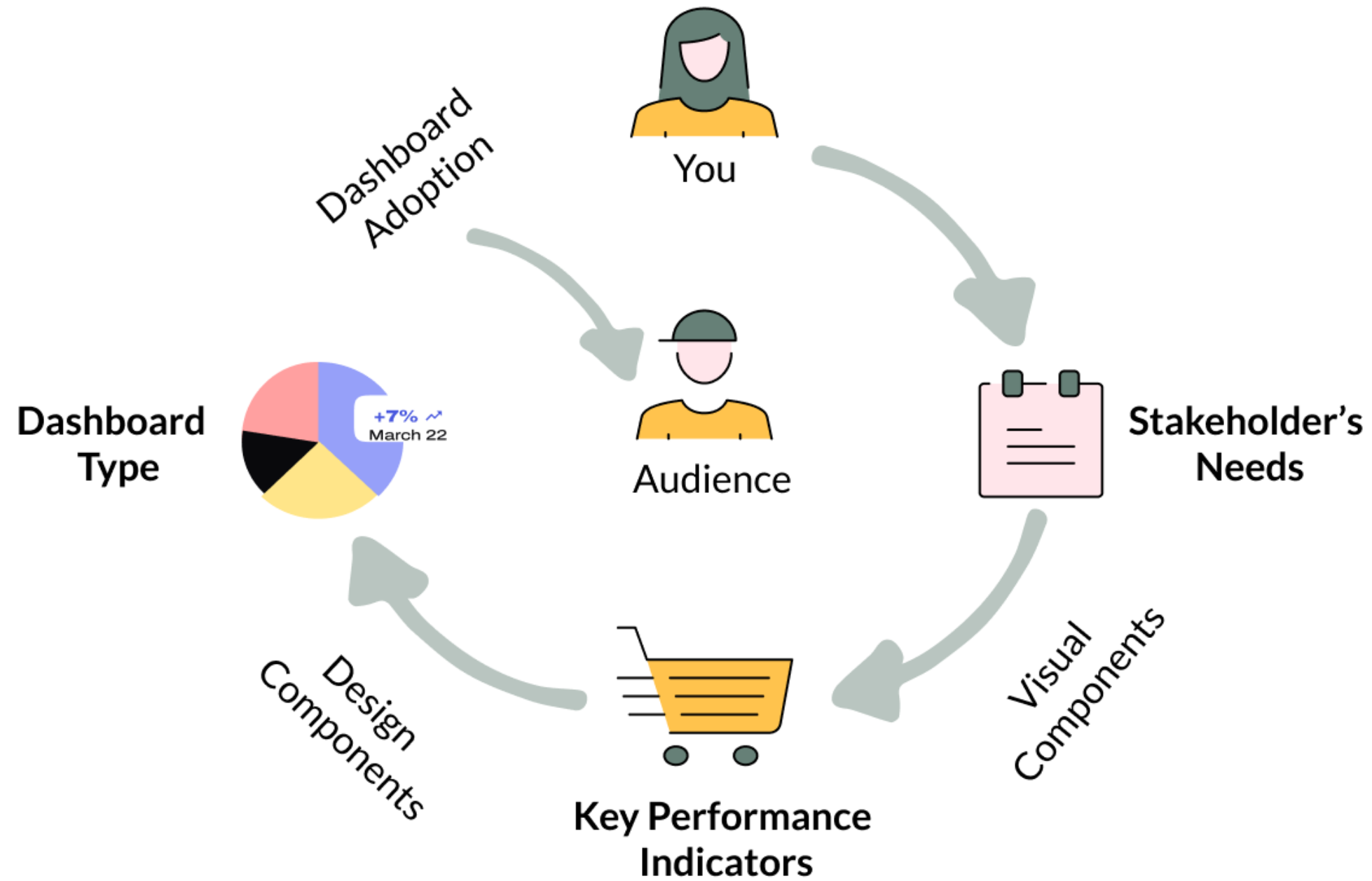
Course overview



Course overview



Course overview



Let's practice!
DASHBOARD DESIGN CONCEPTS

Key performance indicators

DASHBOARD DESIGN CONCEPTS



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President, Scrivner Solutions Inc

What are key performance indicators?

"If you can't measure it, you can't manage it"



Goal



Key performance indicators

- Monitor the company's health
- Track progress against business goals
- Inform strategic and day-to-day decision making
- Highlight areas that need attention
- Provide a basis for strategic and operational improvement

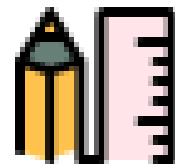
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Goal **Measurement**



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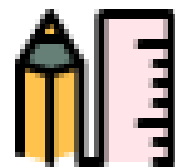
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Goal



Measurement



Frequency



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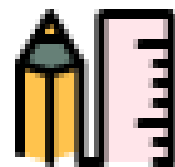
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Goal



Measurement



Frequency



2 - 4 KPIs

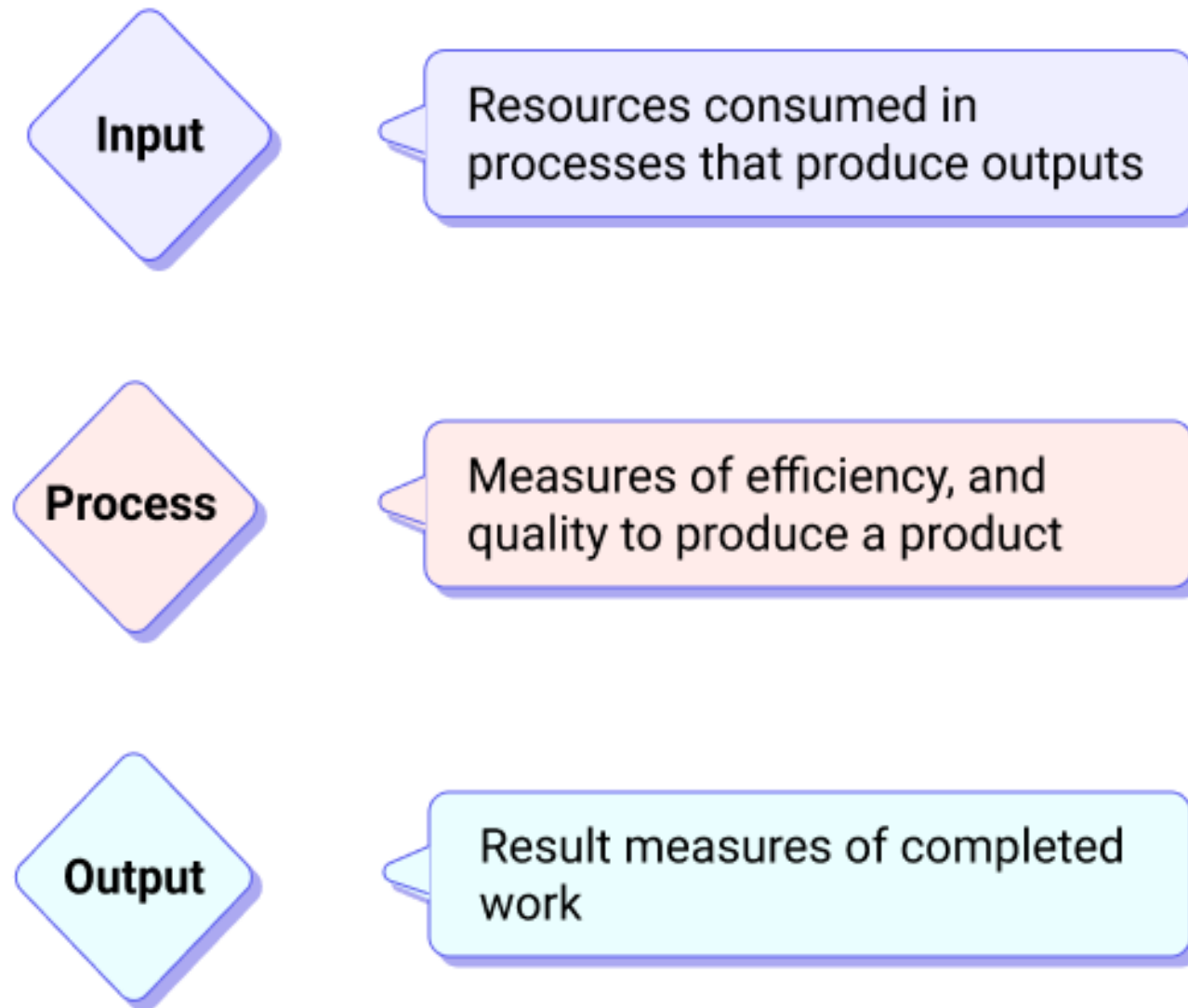
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KPIs measurements and types

Operational measurements



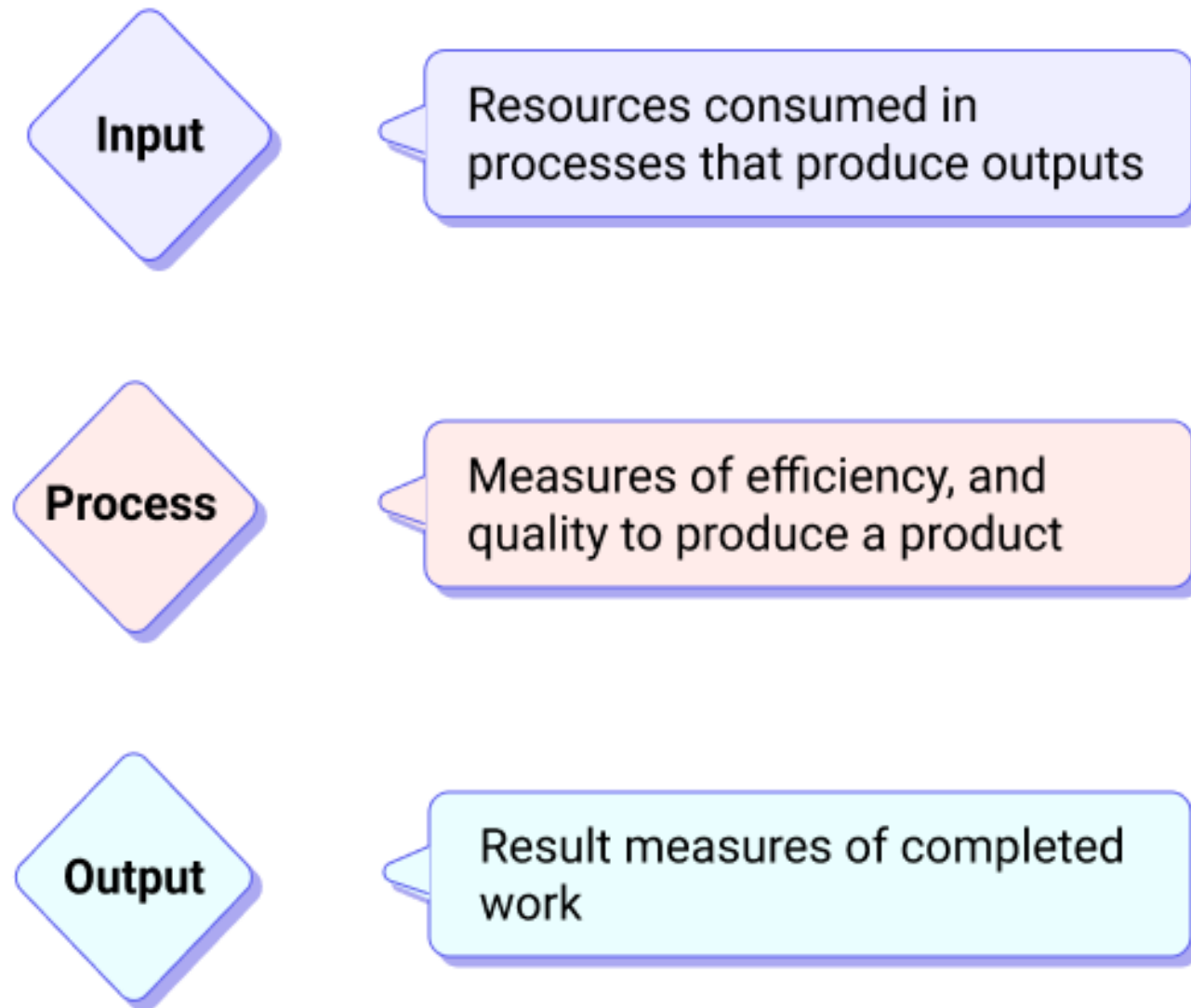
Strategic impact measurements



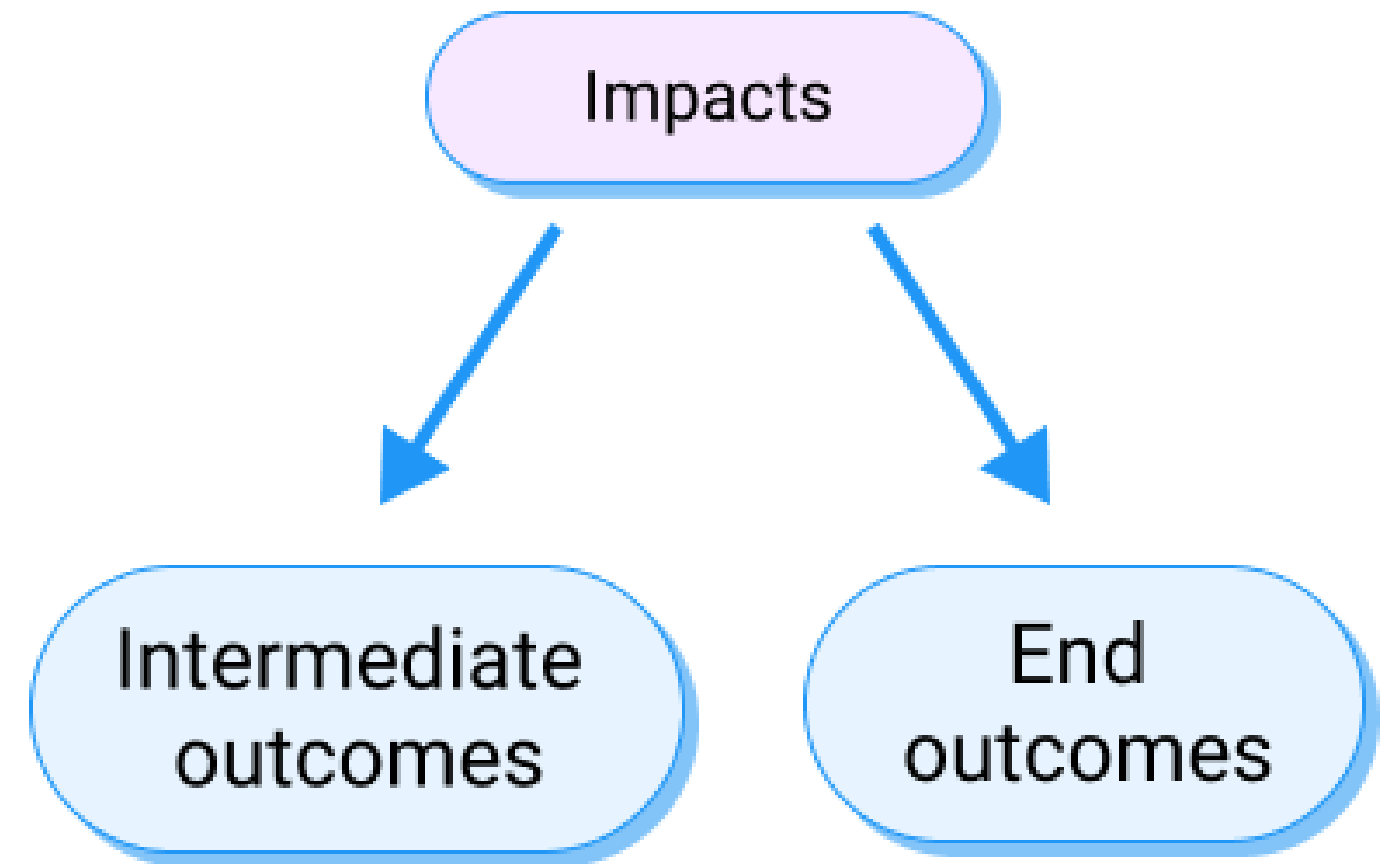
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KPIs measurements and types

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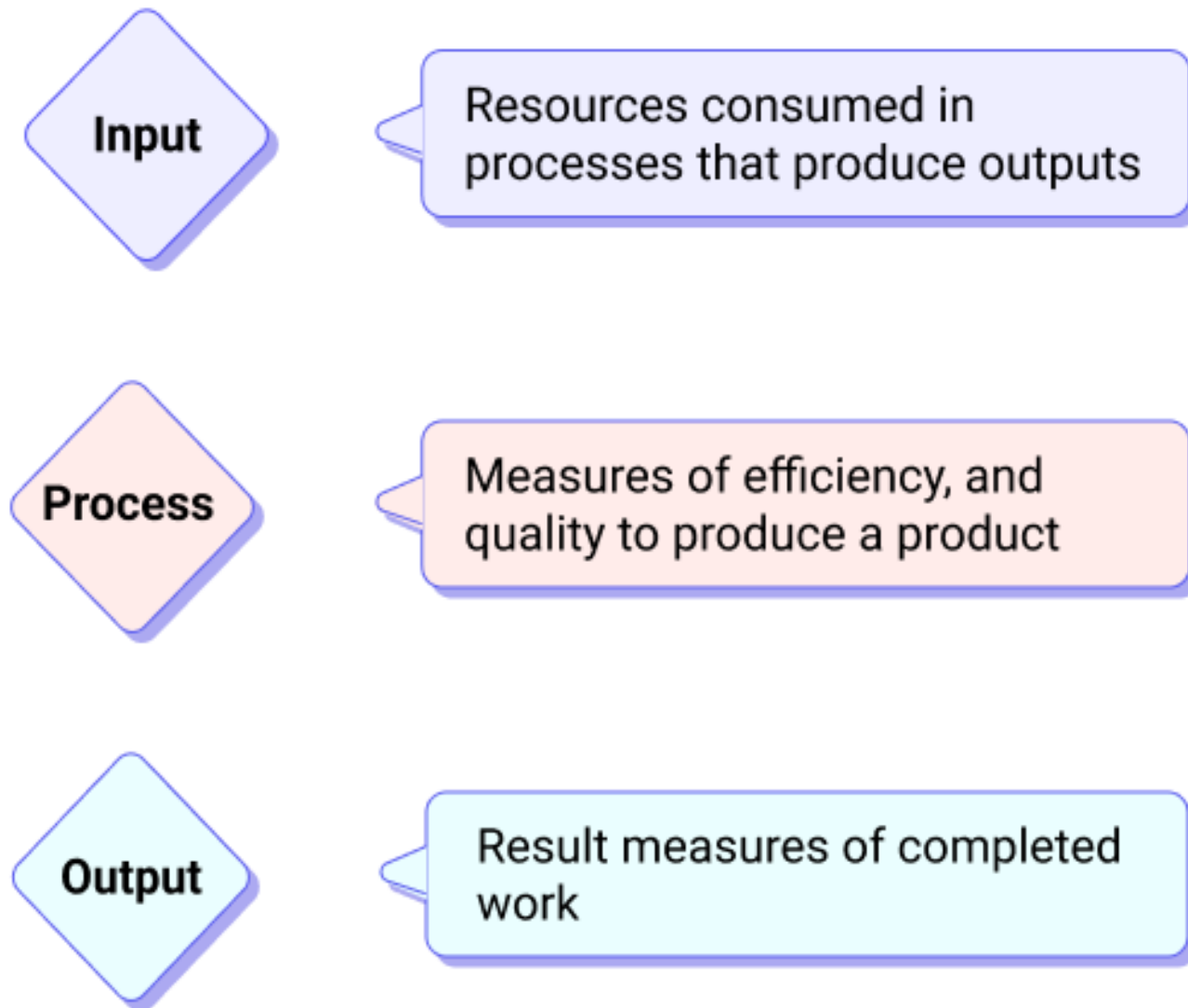
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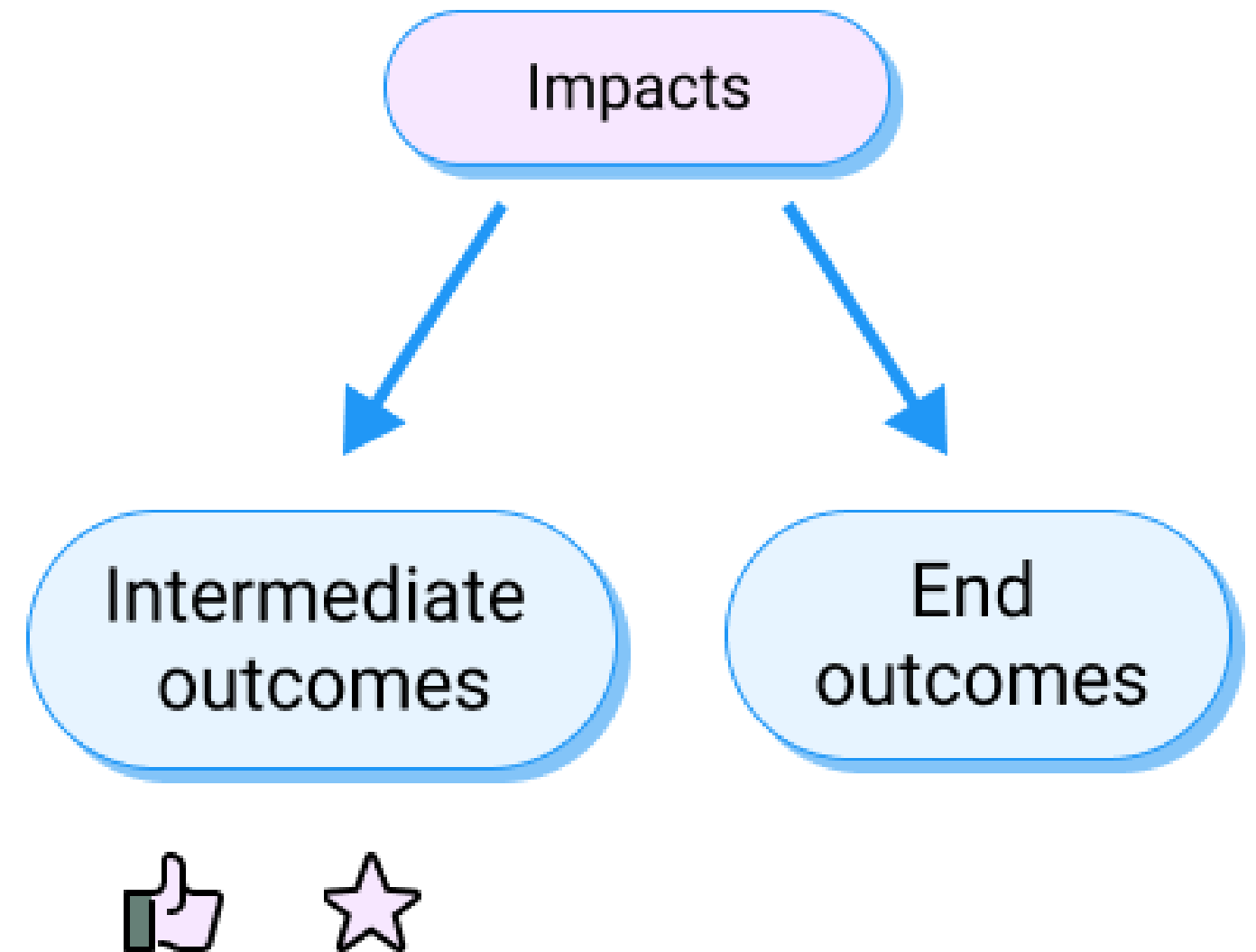
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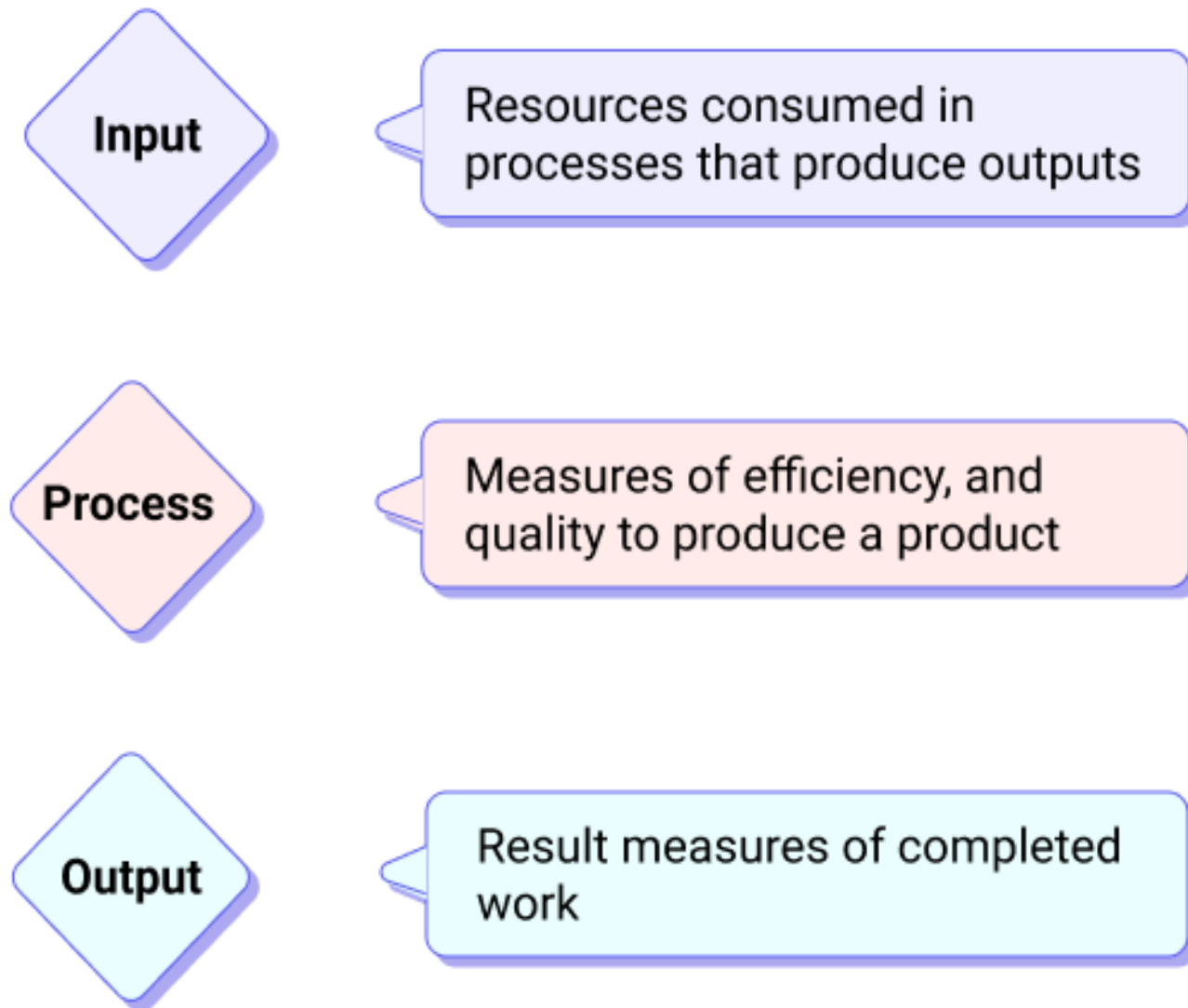
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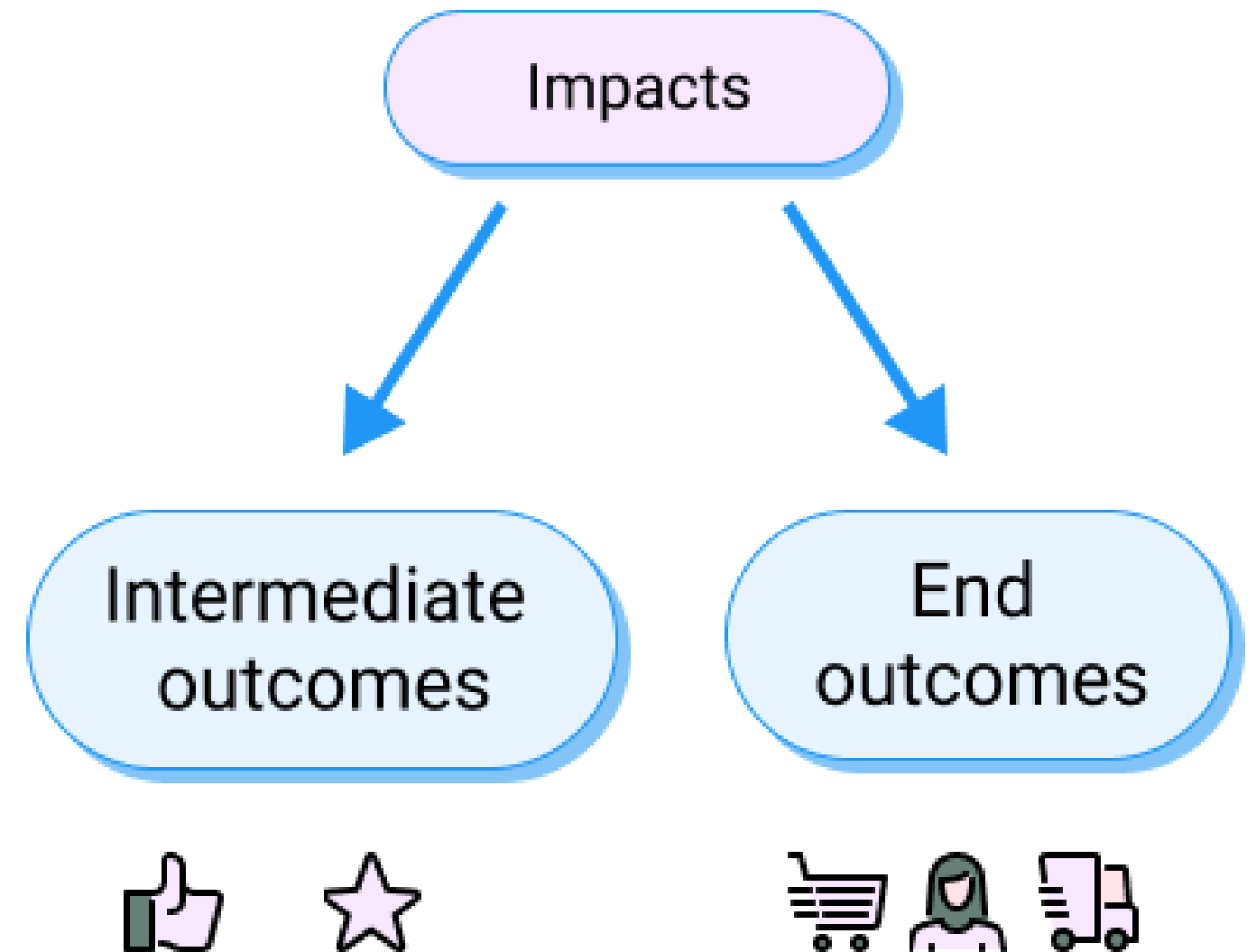
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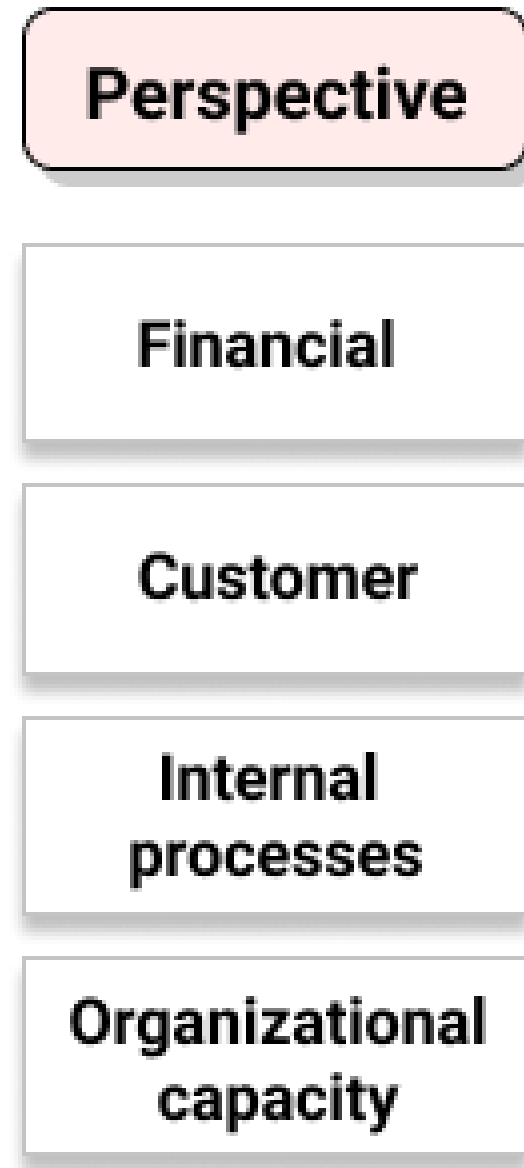
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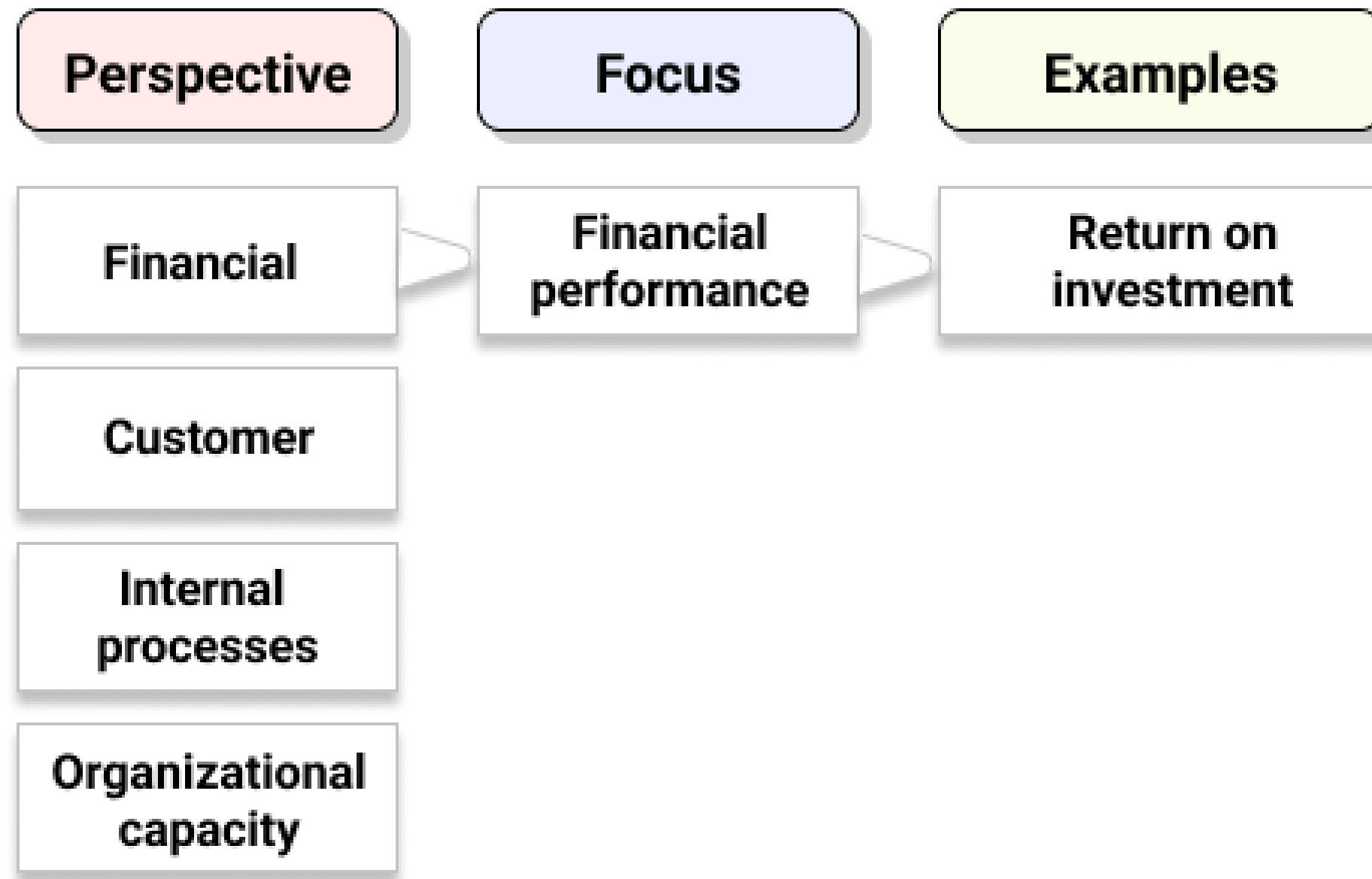
Kaplan & Norton balanced KPIs

Effective business management: KPIs from different perspectives



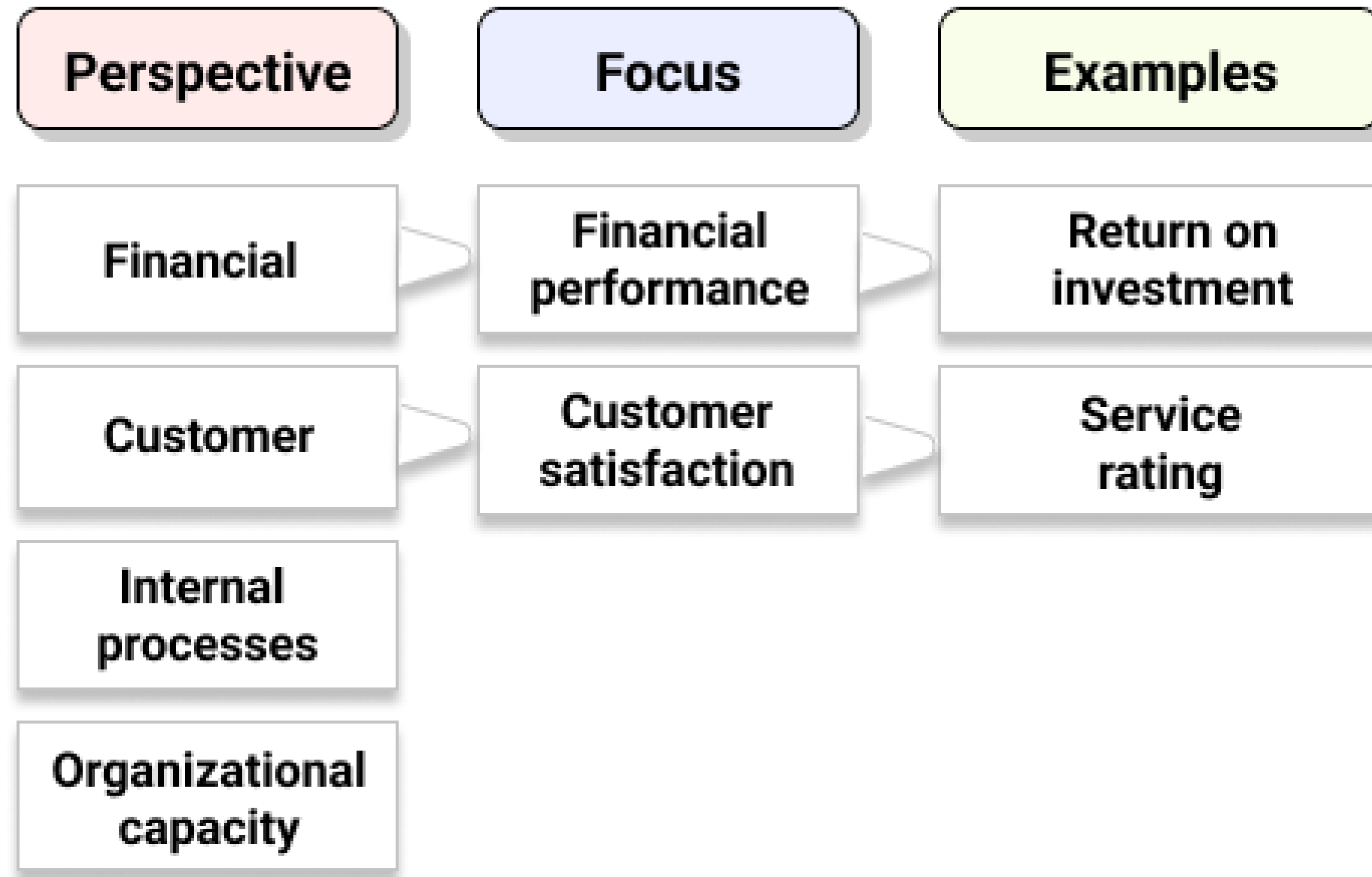
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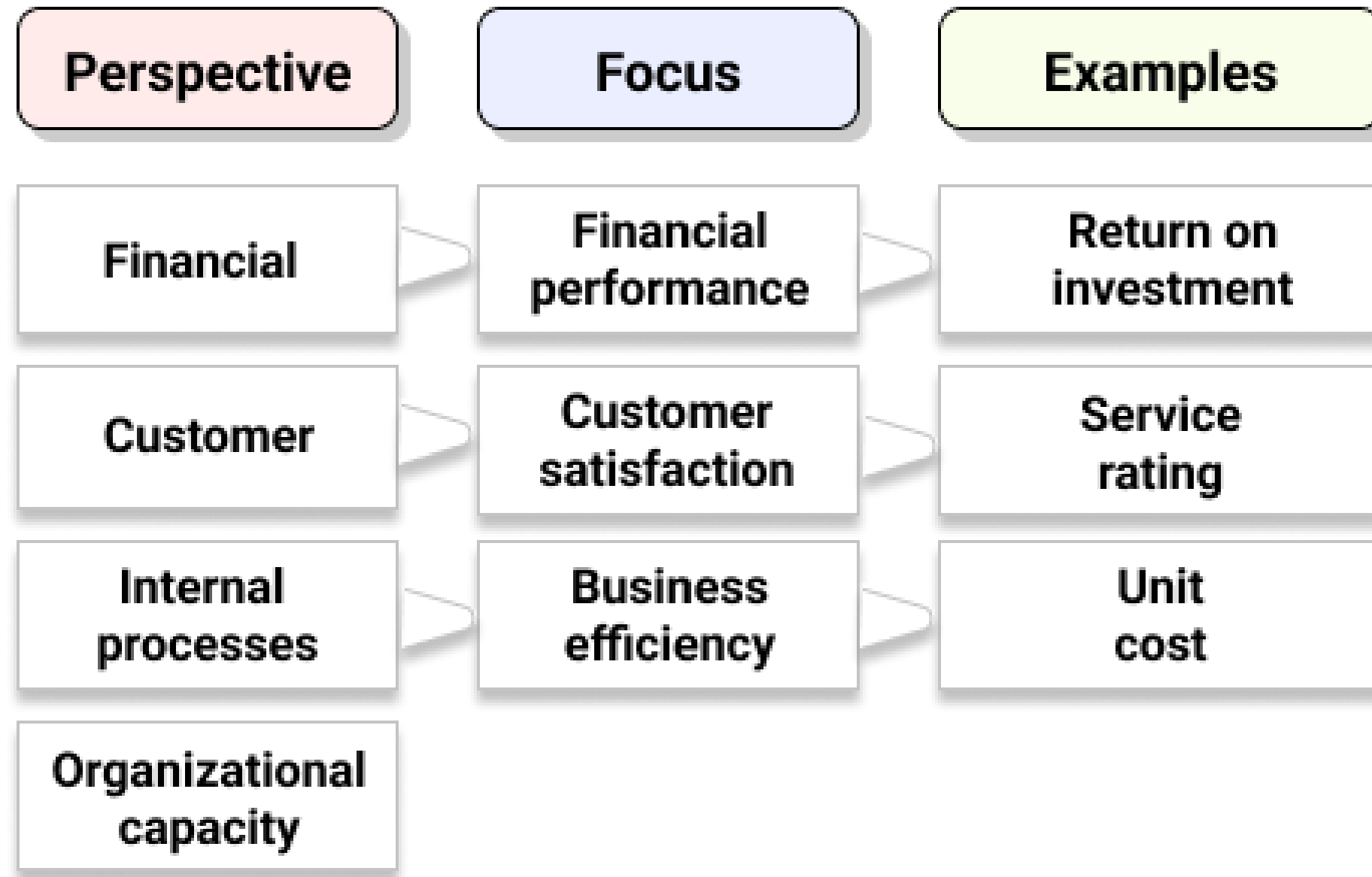
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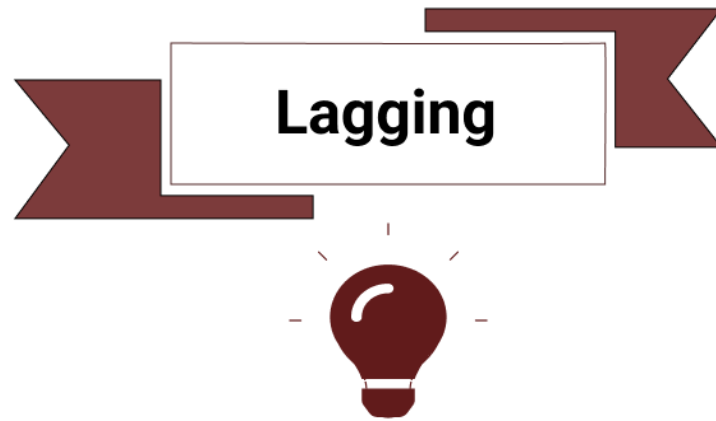


Kaplan & Norton balanced KPIs

Effective business management: KPIs from different perspectives

Perspective	Focus	Examples
Financial	Financial performance	Return on investment
Customer	Customer satisfaction	Service rating
Internal processes	Business efficiency	Unit cost
Organizational capacity	Knowledge & innovation	Employee retention

Are you leading or lagging?



- *Where your business has been*
- Outcomes (output metrics)
- High-level metrics (revenue)
- Example: sales revenue

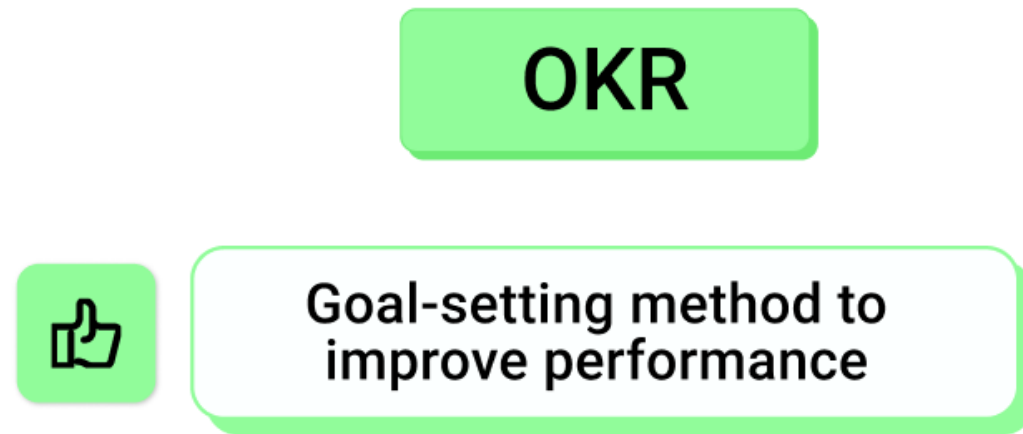
- *Where your business is going*
- Progress (input metrics)
- Faster feedback
- Example: number of calls per sale rep



¹ Watts, Stephen. (2019) "Leading vs. Lagging Indicators: What's the Difference?"

Objectives and key results (OKRs)

Objectives: significant/defined goal + **Key Results:** measurable success criteria to track goals



¹ Maasik, Alexander. 2021. Step-by-step guide to OKRs. Weekdone.

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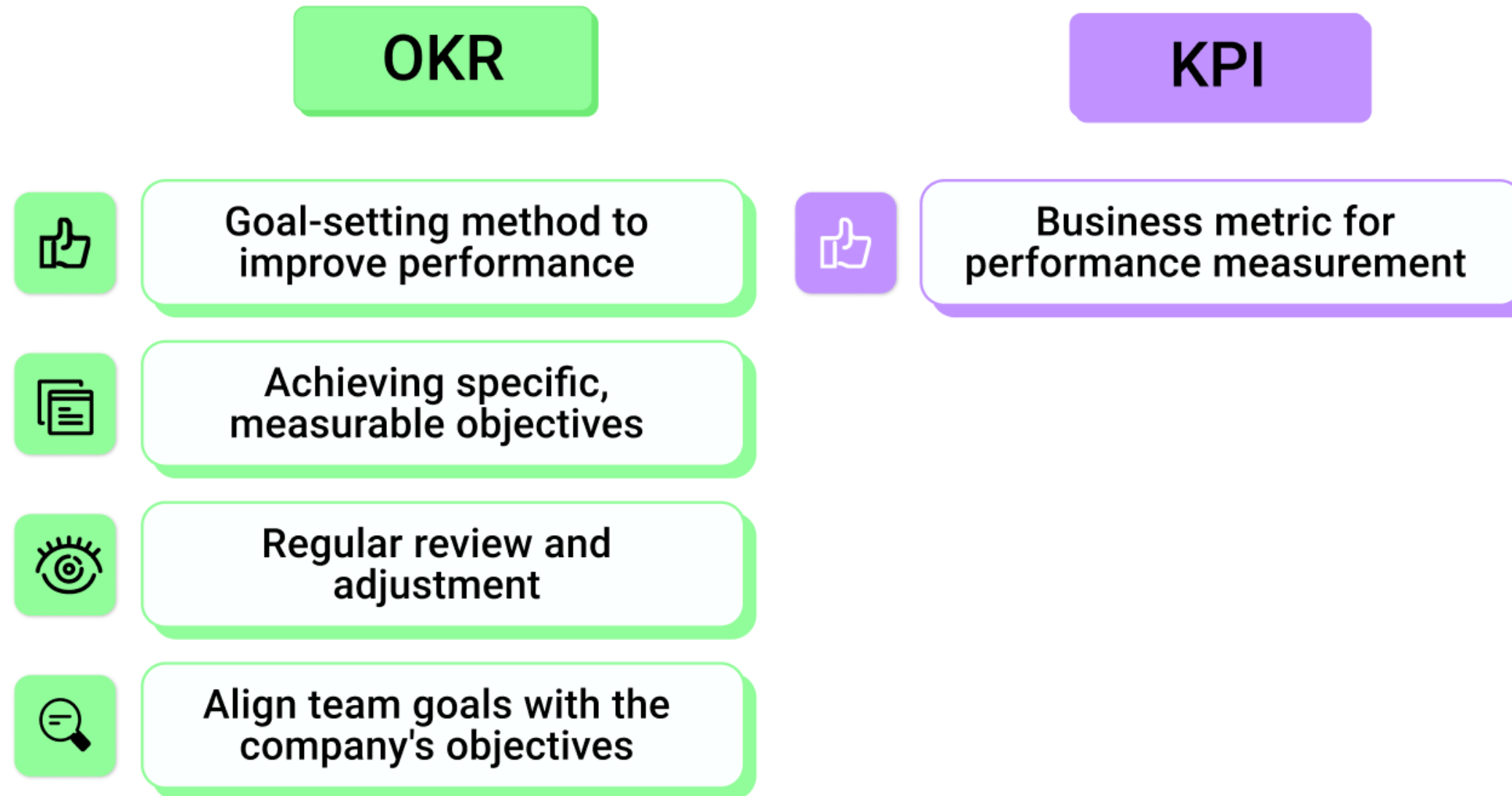
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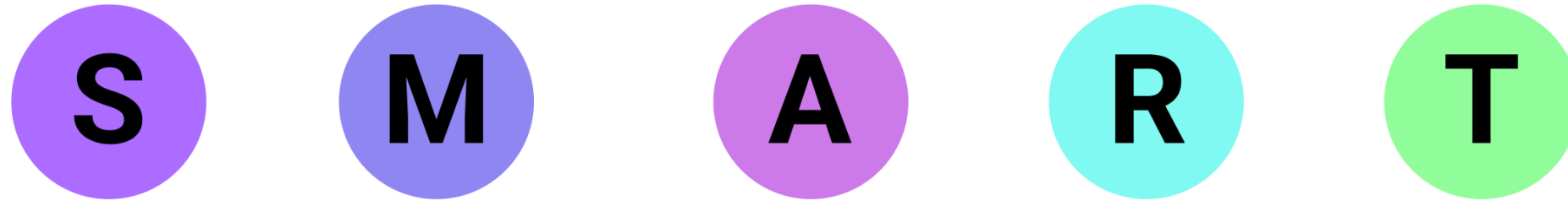
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OKR		KPI	
	Goal-setting method to improve performance		Business metric for performance measurement
	Achieving specific, measurable objectives		Progress towards pre-determined targets
	Regular review and adjustment		Re-evaluated on an annual basis
	Align team goals with the company's objectives		Often used by leadership

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Goals: be S.M.A.R.T. or even S.M.A.R.T.E.R.

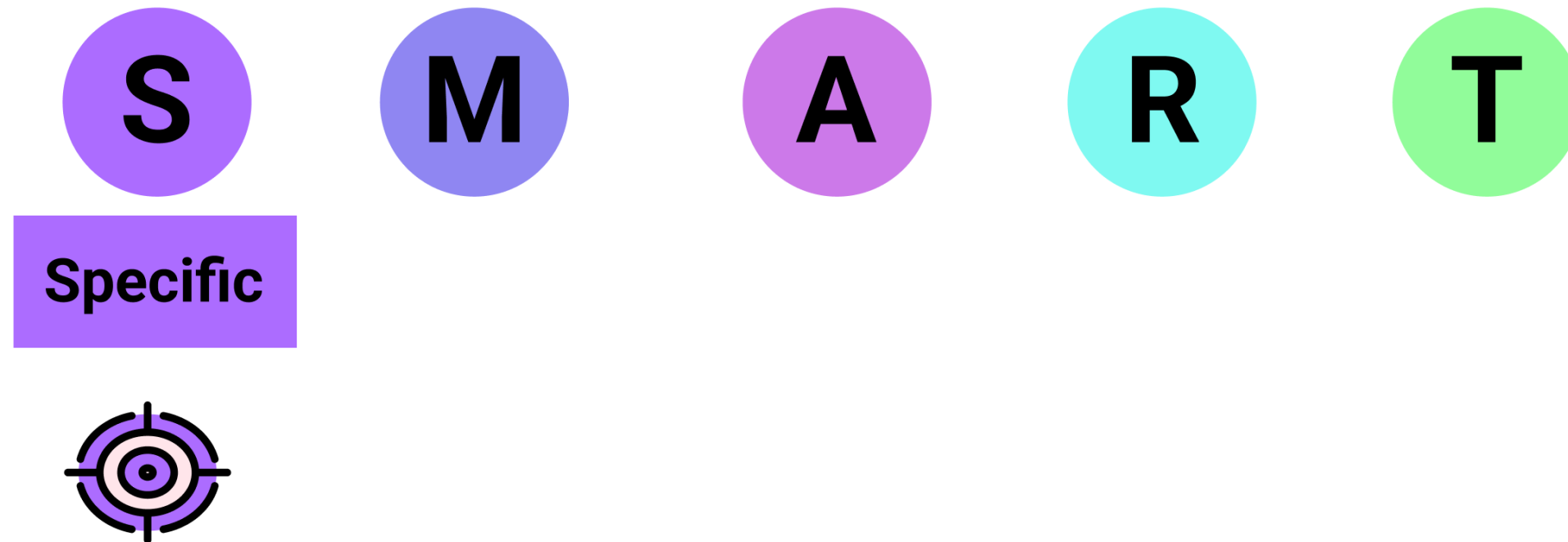
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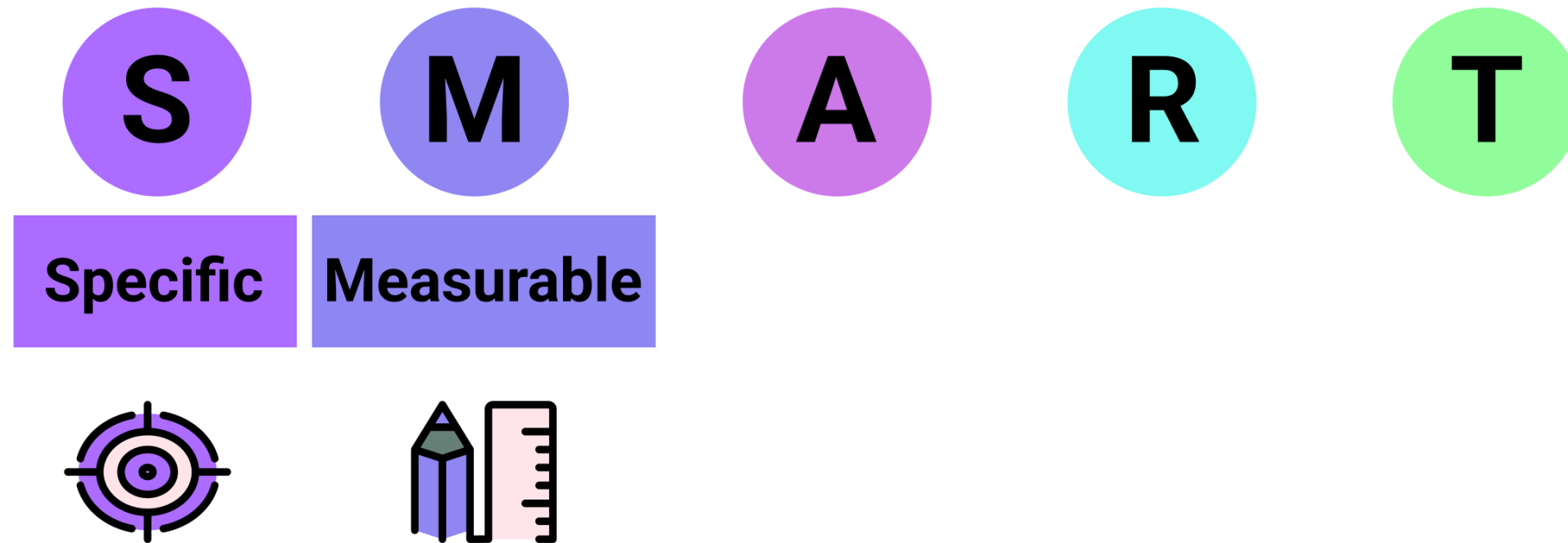
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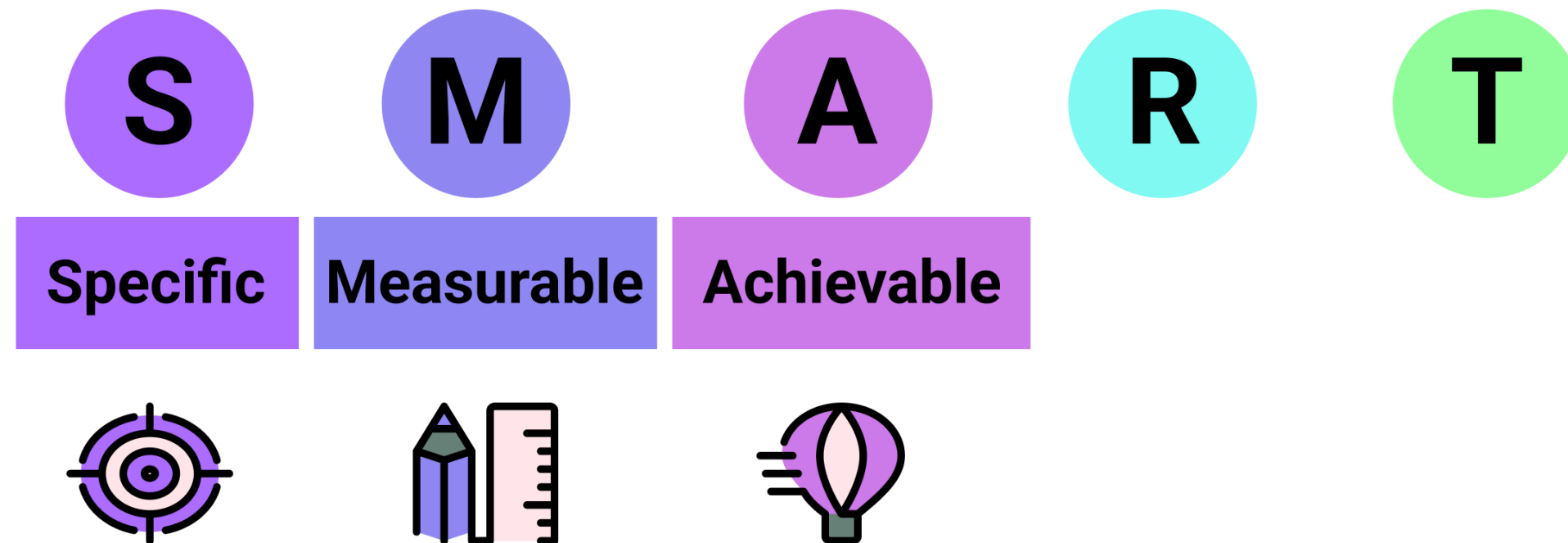
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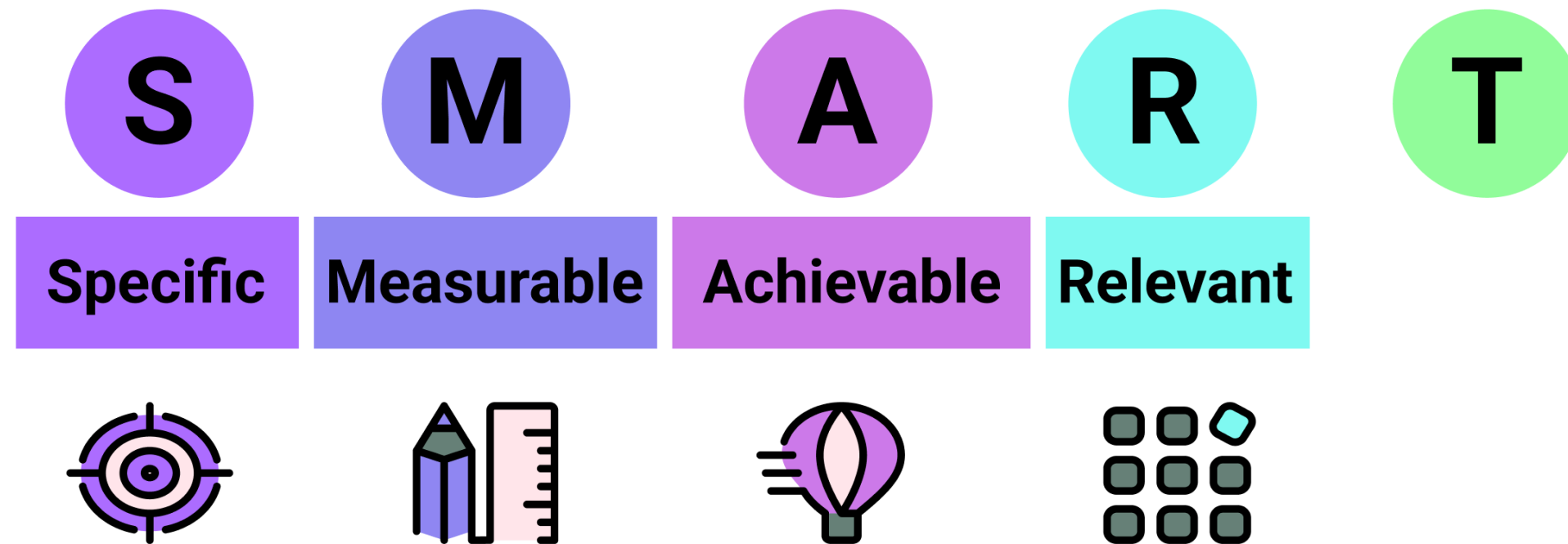
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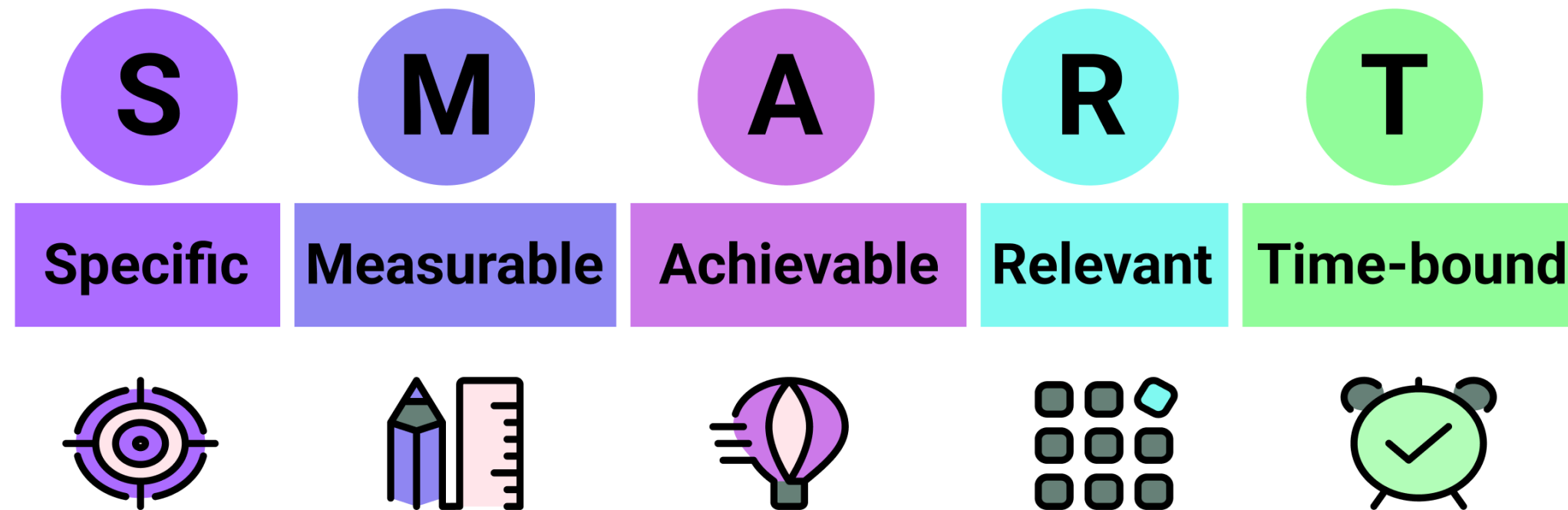
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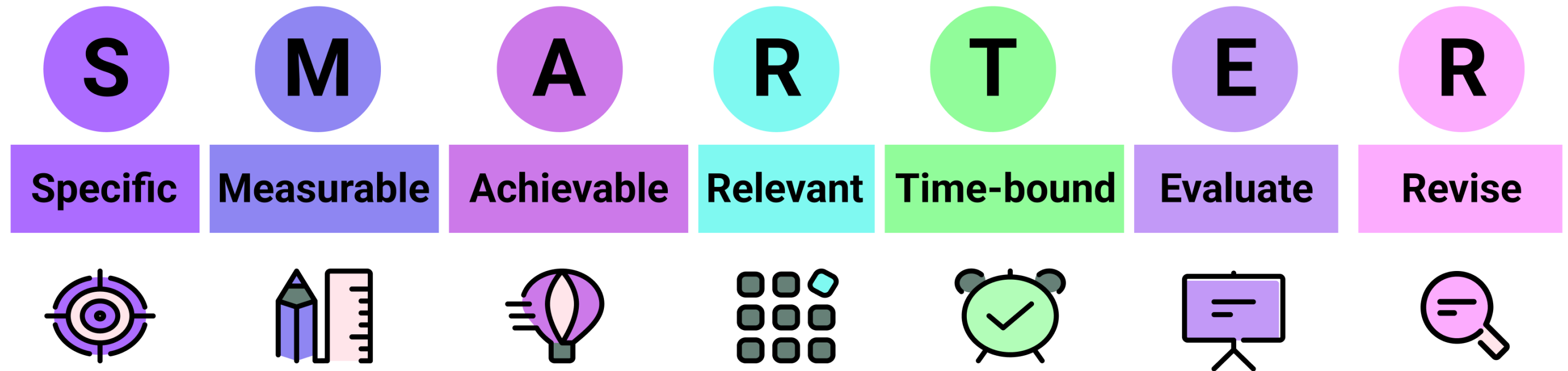
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Let's practice!
DASHBOARD DESIGN CONCEPTS

Stakeholders

DASHBOARD DESIGN CONCEPTS



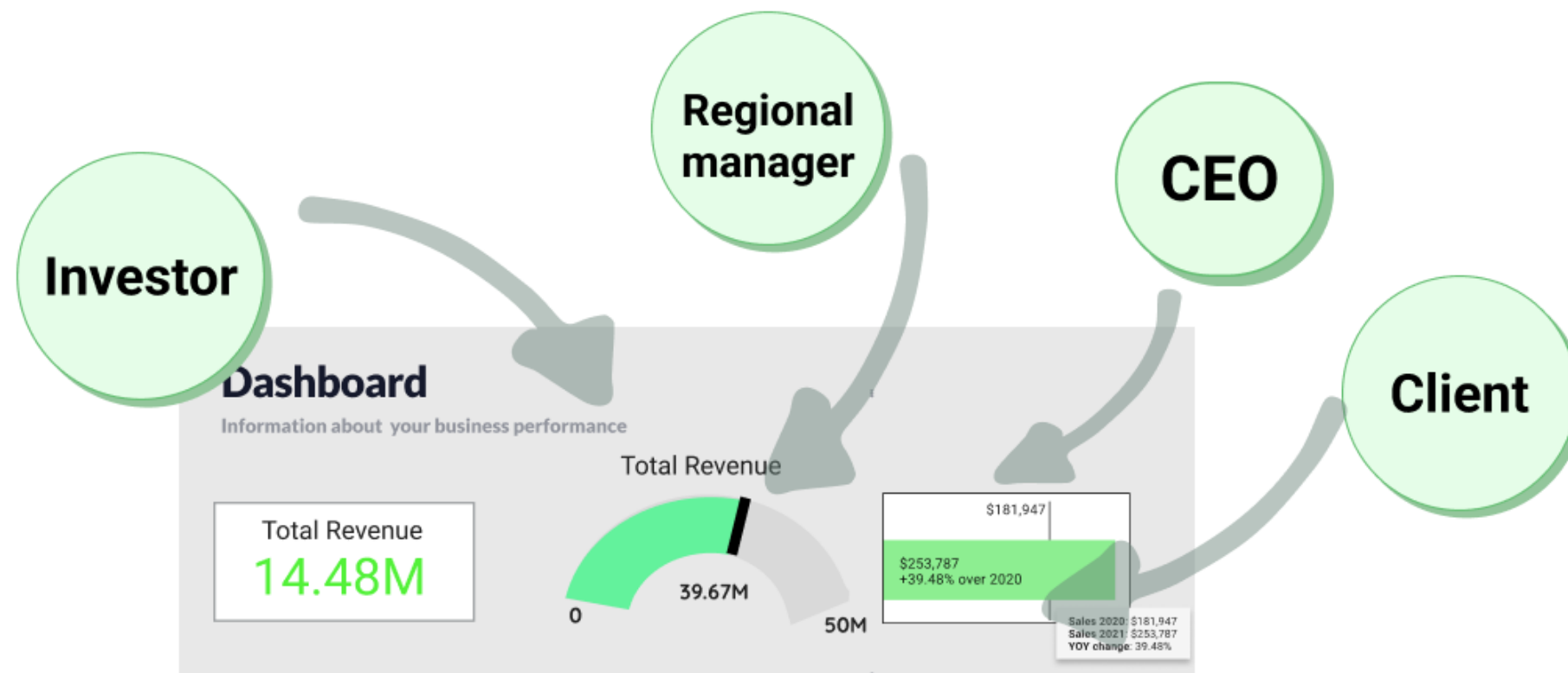
Olga Scrivner, PhD

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Dashboard design and stakeholders

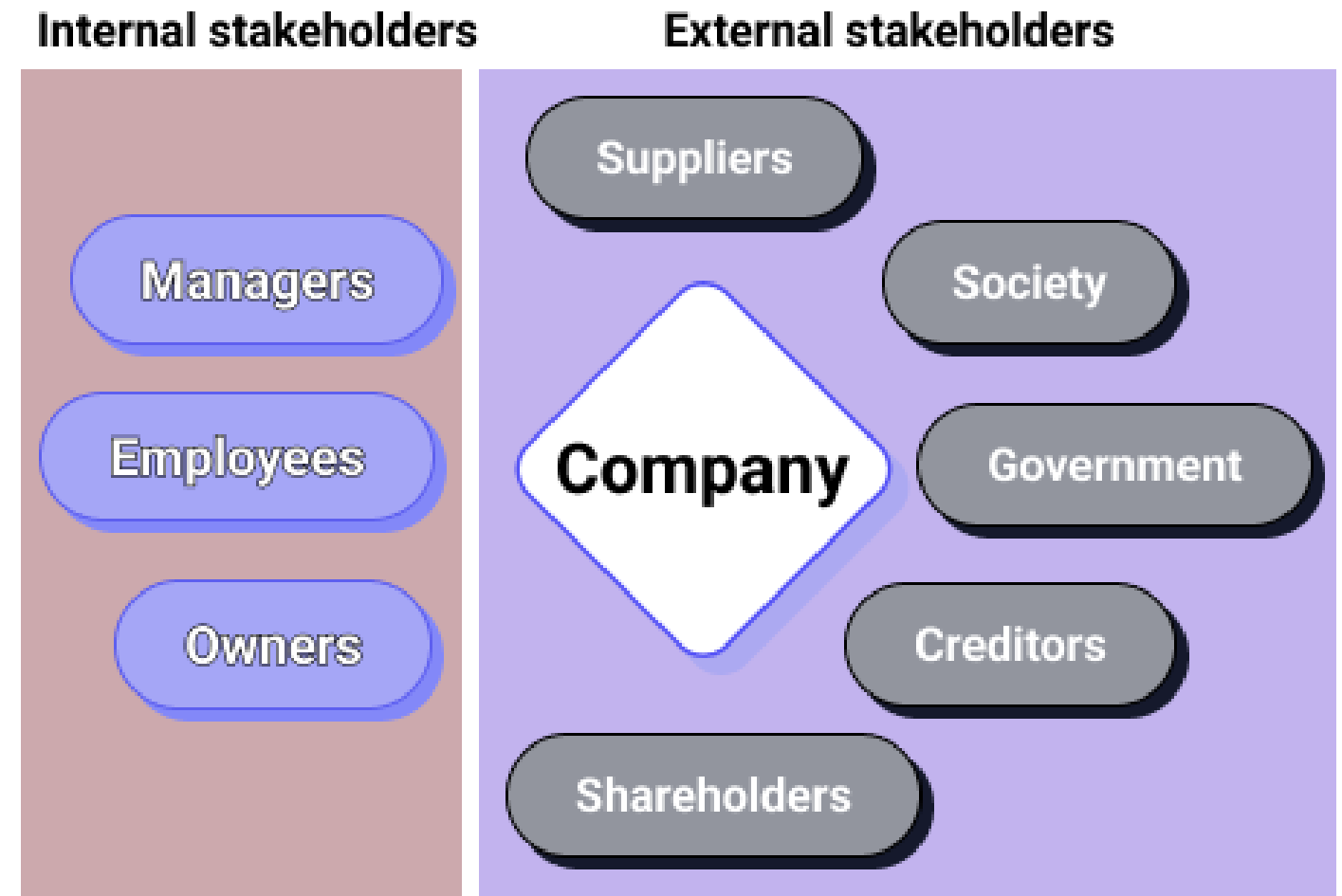
A dashboard is viewed by people with different roles, responsibilities and levels of expertise.

- Rarely do two people have the same exact goals
- Rarely do two people require the same exact insights
- Understanding stakeholders and their needs will increase dashboard effectiveness



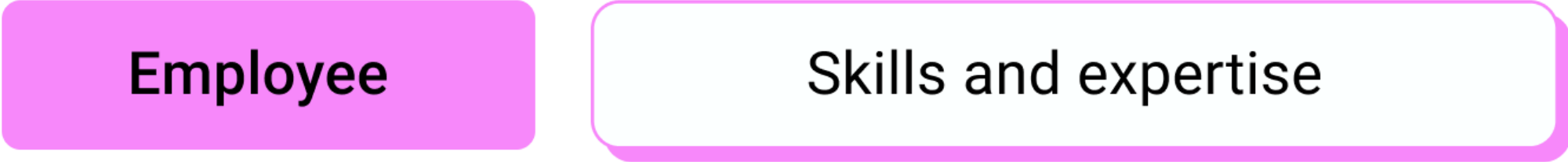
Who are the stakeholders?

- **Key stakeholders** help companies make strategic decisions
- **Internal stakeholder:** the interest comes through a direct relationship, such as employment and ownership
- **External stakeholders:** the impact from the business actions and outcomes



Stakeholder's contributions

Stakeholders provide specific and different attributes to an organization, however they each have a "stake" in the success of the organization.



Employee

The diagram consists of two rounded rectangular boxes. The first box on the left is solid magenta and contains the text 'Employee'. The second box on the right is white with a magenta border and contains the text 'Skills and expertise'. The boxes are positioned side-by-side, suggesting a direct relationship between the stakeholder and their contribution.

Skills and expertise

Stakeholder's contributions

Stakeholders provide specific and different attributes to an organization, however they each have a "stake" in the success of the organization.

Employee

Skills and expertise

Shareholders

Money and capital

Stakeholder's contributions

Stakeholders provide specific and different attributes to an organization, however they each have a "stake" in the success of the organization.

Employee

Skills and expertise

Shareholders

Money and capital

Customers

Revenue from purchase

Stakeholder's contributions

Stakeholders provide specific and different attributes to an organization, however they each have a "stake" in the success of the organization.

Employee	Skills and expertise
Shareholders	Money and capital
Customers	Revenue from purchase
Public	Customer loyalty

Dashboard design stages

Define

Who the dashboard is for and what metrics are critical

¹ Davis, Matt. (2021). Identifying Key Roles.

Dashboard design stages

Define

Who the dashboard is for and what metrics are critical

Prototype

How to display critical metrics

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Dashboard design stages

Define

Who the dashboard is for and what metrics are critical

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Build

Adding real data to the dashboard

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Dashboard design stages

Define

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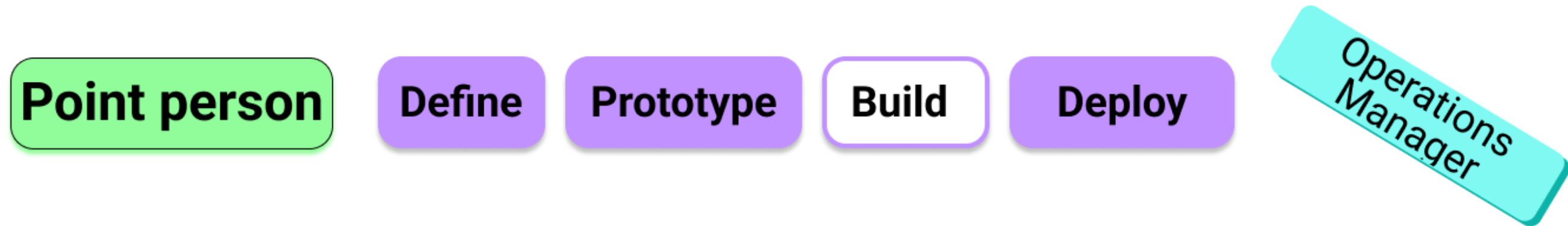
Adding real data to the dashboard

Deploy

Sharing the dashboard with audience

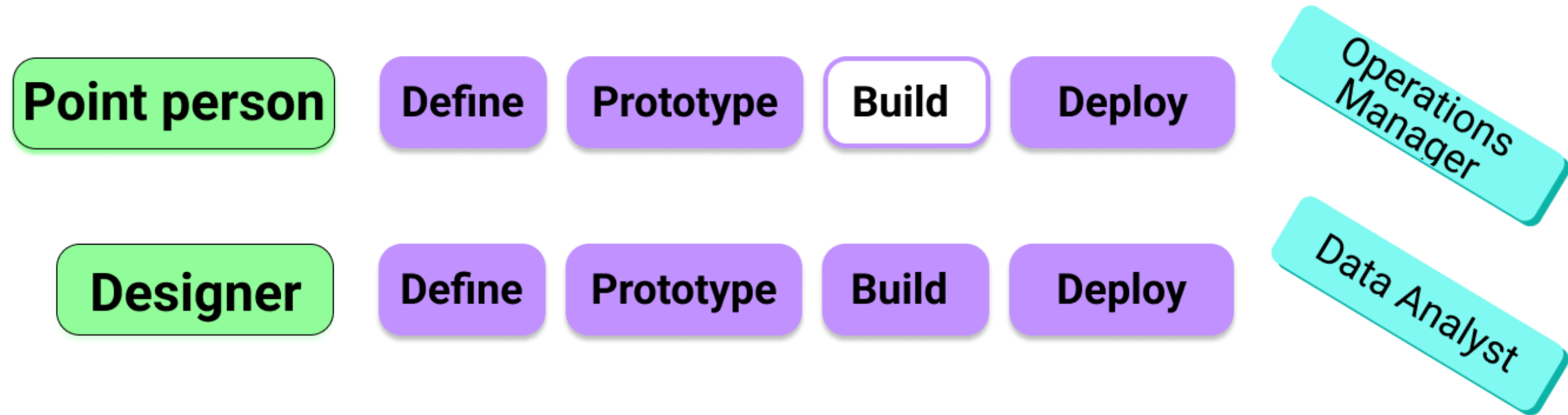
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Dashboard design key stakeholders



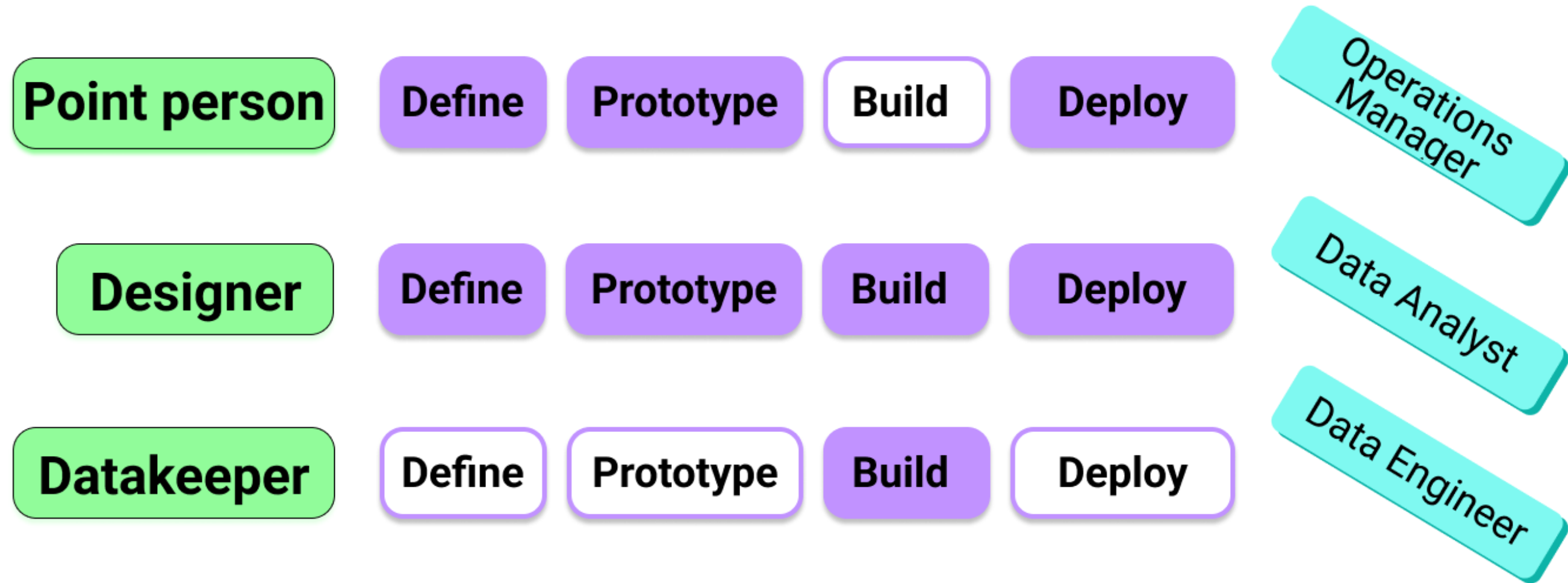
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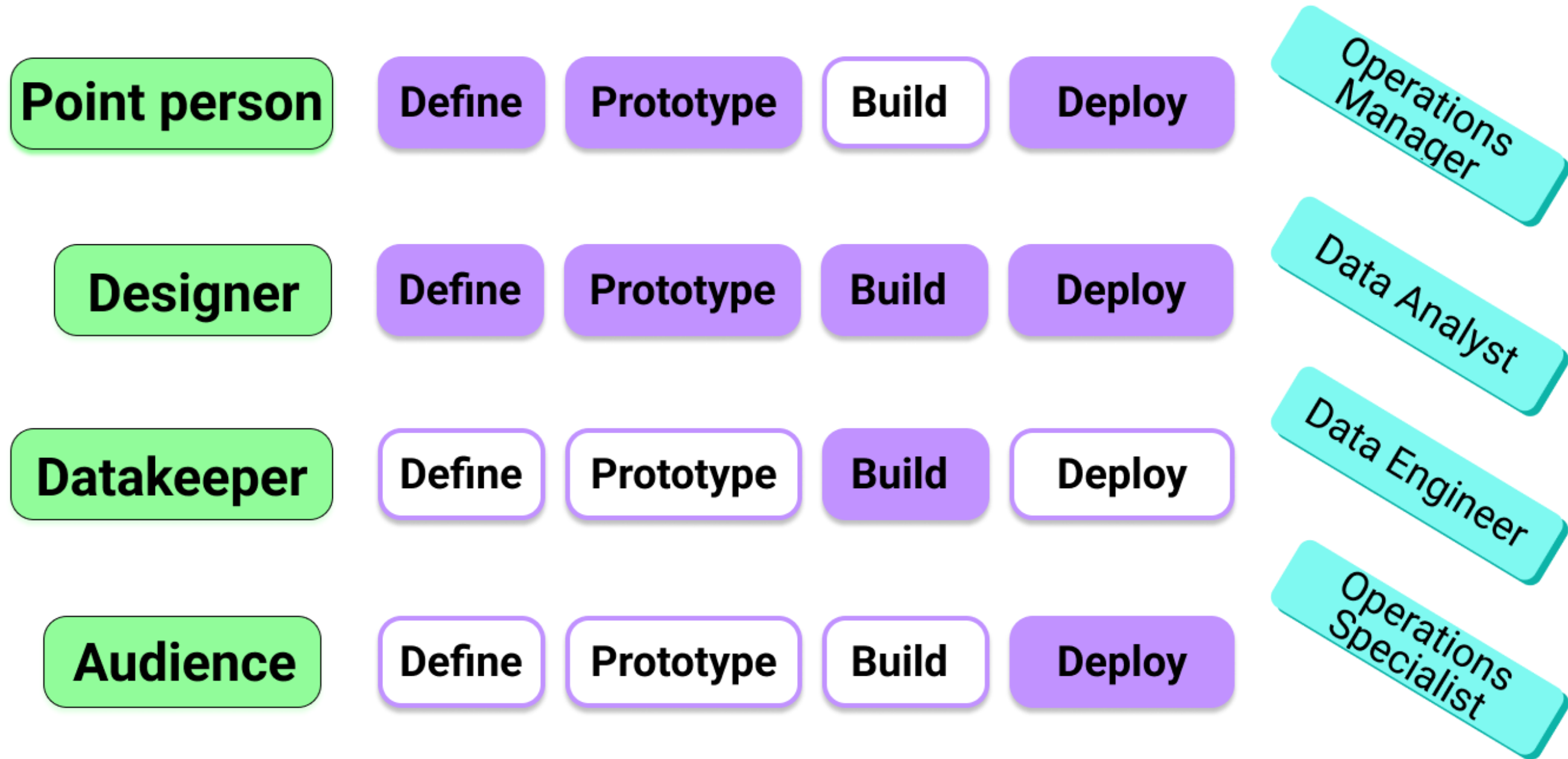
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Dashboard design key stakeholders



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Dashboard design key stakeholders



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Identifying stakeholders' responsibilities



Identify key metrics, audience

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Identifying stakeholders' responsibilities



Identify key metrics, audience



Define stakeholders,
metrics calculations

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Identifying stakeholders' responsibilities



Identify key metrics, audience



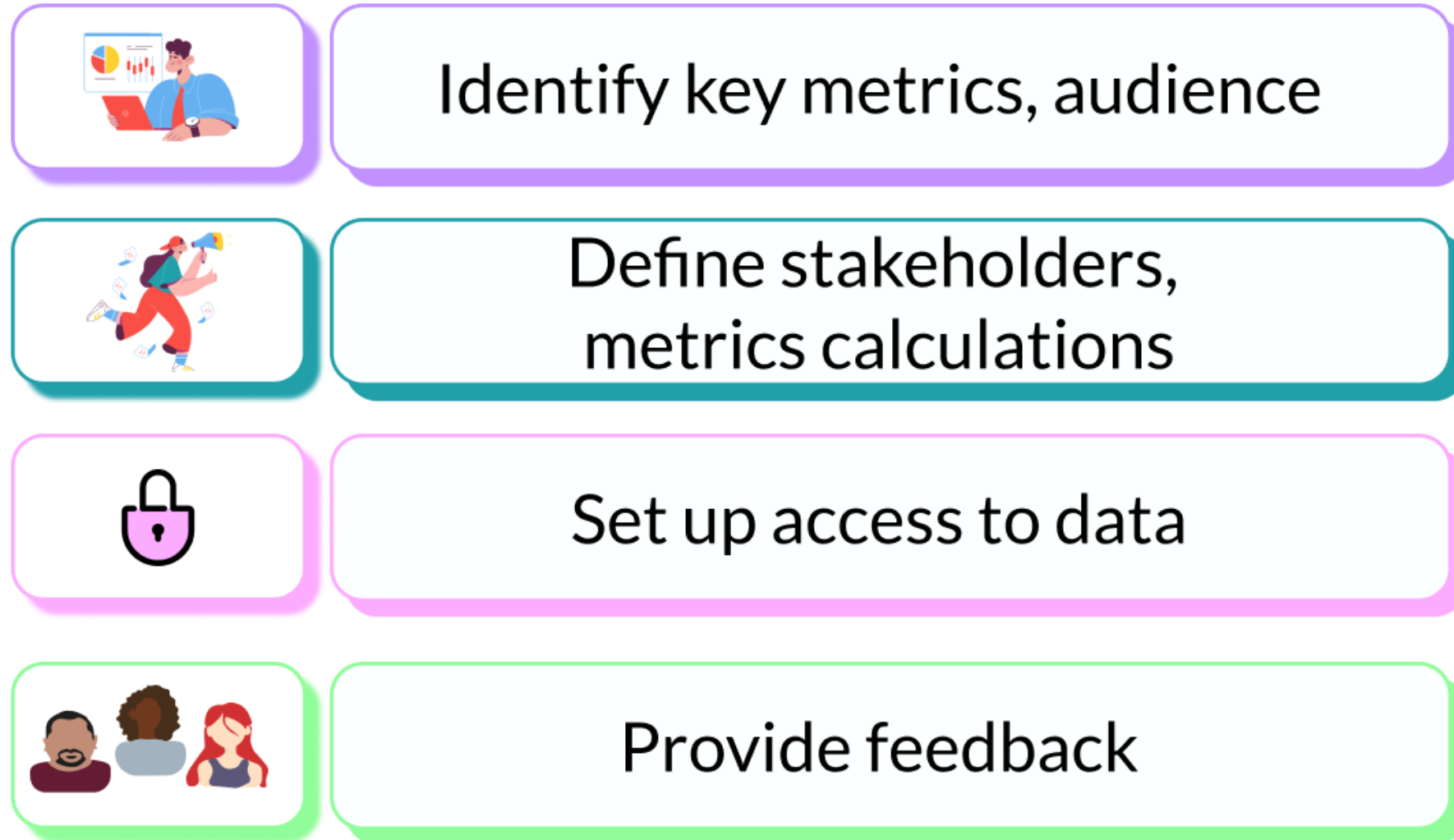
Define stakeholders,
metrics calculations



Set up access to data

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Identifying stakeholders' responsibilities



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Conducting an interview with stakeholders

Question 1

Why do you need a dashboard?
What do you hope to achieve?

¹ Kalaivanan, Prague. How Understanding Stakeholders Increases Dashboard Effectiveness.

Conducting an interview with stakeholders

Question 1

Why do you need a dashboard?
What do you hope to achieve?

Question 2

What are the essential KPIs needed
to make the dashboard effective?

¹ Kalaivanan, Prague. How Understanding Stakeholders Increases Dashboard Effectiveness.

Conducting an interview with stakeholders

Question 1

Why do you need a dashboard?
What do you hope to achieve?

Question 2

What are the essential KPIs needed
to make the dashboard effective?

Question 3

Where and how will
the dashboard be used?

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Question 1

Why do you need a dashboard?
What do you hope to achieve?

Question 2

What are the essential KPIs needed
to make the dashboard effective?

Question 3

Where and how will
the dashboard be used?

Question 4

What are the company's strategic
objectives?

¹ Kalaivanan, Prague. How Understanding Stakeholders Increases Dashboard Effectiveness.

Let's practice!
DASHBOARD DESIGN CONCEPTS